Network Differentiation and Subscription Pricing in Electric Vehicle Fast Charging

John Higgins[*]

Department of Economics
University of Wisconsin-Madison[†]

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Abstract

This paper studies the interaction between market power and price discrimination in the German electric vehicle (EV) fast charging market. Large charging networks use two-part tariffs in the form of subscription plans to price discriminate based on consumers' expected charging needs. Using novel, high-frequency fast charging data, rich consumer microdata, and large-scale commuting and mobility data, I estimate a structural discrete choice model of spatial charging demand. Consumers are heterogeneous in their routes, alternative charging options, and preferences for charging speed. Consumers choose whether to subscribe to a plan based on their expected charging needs. Using my model, I conduct counterfactual scenarios to decompose the effect of network differentiation and subscription pricing on market power. I find that eliminating price discrimination causes prices to decrease by 9.0% on average, while aggregate demand increases by 5.5%. Demand reallocates from networks which offered subscriptions to those that did not. I also find that network size enhances firms' ability to price discriminate, allowing networks to capitalize on advantageous positions.

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[†] Email: jfhiggins@wisc.edu

1 Introduction

As of July 2026, electric vehicles (EVs) comprised 29.3% of new car sales in Germany, with EVs representing 4.1% of the current vehicle fleet (KBA 2026). The German government has committed billions of euros in subsidies to accelerate the EV transition, and the European Union has sought a 90% reduction of tailpipe CO_2 emissions by 2035 (Reuters 2025). Fast charging infrastructure - which enables charging in minutes, rather than hours - is a critical component of this transition. Fast charging enables en route charging and mitigates the fear of running out of charge before reaching one's destination (ACEA 2024). In order for fast charging to effectively accelerate the EV transition, it must be both accessible and affordable.

I study the German market for public fast charging. Charge point operators (CPOs) sell charging through networks of stations, and almost all of them set a single price per kWh across their entire network. CPOs therefore compete on two margins: the size and location of their networks, and the menus of prices they offer. On both margins, a few large networks stand apart. EnBW and Ionity were among the few firms allowed to build stations at Autobahn service areas, an allocation that a German court ruled illegal in 2026.[1]. Aral Pulse operates at Autobahn-adjacent truck stops and an extensive network of gas stations. These networks have among the highest utilization rates in the market. These three networks are also alone in offering major subscription contracts.

This paper measures the effect of network concentration on the cost of charging. Although prices do not vary across a network's stations, CPOs compete through nonlinear contracts by offering subscription tiers. A subscription gives heavy users a discount: a monthly fee in exchange for a lower price per kWh. Because a subscription plan is worth more on a network that covers more of a driver's charging needs, differentiation between networks creates scope for market power distortions. Under monopoly, second-degree price discrimination raises the unit price paid by low-usage consumers above marginal cost (Mussa and Rosen 1978, Maskin and Riley 1984). Competition between networks can eliminate this distortion. Firms then compete for heavy users, and subscription pricing can raise welfare relative to linear pricing (Armstrong and Vickers 2001). Whether competition removes the distortion depends on how differentiated firms are (Rochet and Lars A. Stole 2002). In EV charging, networks are differentiated by the size, location, and characteristics of their stations. Consumers differ in how much they value this differentiation.

My objective is twofold. First, I quantify the competitive distortions caused by subscription pricing relative to linear pricing. Second, I measure how network differentiation determines the ability of firms to price discriminate and earn market power rents. To do so, I estimate a spatial model of demand for fast charging in the style of Berry, Levinsohn, and Pakes 1995 and Houde 2012. Consumers differ in two dimensions that determine their need for public fast charging: their access to home or workplace charging, and the trips they make each month. The length of these trips and the efficiency of the vehicle determines the number of charging events each consumer must complete. On each charging event, the consumer chooses between slow charging (at home, at work, or at a public slow charging, depending on availability) and the fast chargers located along the route. At the beginning of the month, the consumer chooses whether to subscribe to a plan based on the expected utility from charging under each plan. Networks are therefore differentiated by the location of their stations relative to consumers' routes, and the value of this differentiation increases with expected usage. Subscription plans are worth purchasing if one can do enough charging in-network to justify the fee. The more a user charges in-network, and the more convenient that network is to access for that consumer, the higher their willingness to pay for the plan will be. This creates a complementarity between expected usage and network differentiation.

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[1] The German Monopoly Commission argued that this allocation limited competition along the Autobahn. Tesla and Fastned challenged it in court in 2022. Future service area sites must now be allocated through a competitive tender

A main contribution of this paper is a new dataset that measures both the demand for fast charging and the heterogeneity of the drivers who use it. I construct a novel dataset of real-time utilization for the vast majority of fast charging stations in Germany. Between November 2023 and November 2025, I scraped the occupancy of every station that publishes real-time information through the Open Charge Point Interface at three-minute intervals. I recover individual charging events from changes in the number of available charging points. Publicly available data on fast-charging demand are scarce, and these data measure demand station-by-station across an entire national market. These data complement recent studies of the demand for charging stations (Dorsey, Langer, and McRae 2025, Barwick, S. Li, and Xia 2026).

I combine the charging data with three sources that describe where drivers travel and how they charge. A survey of about 1,900 German EV drivers reports access to home and workplace charging, annual mileage, vehicle characteristics, and the share of charging done at each option. Administrative commuting flows between all pairs of German municipalities and a national travel-diary survey of about one million trips determine the routes that drivers take. I map these routes onto the road network to construct each driver's set of fast chargers. Finally, vehicle registrations by zipcode locate EV drivers and measure the distribution of battery sizes and charging speeds. I estimate the model for September 2024 in five large regions, which account for about half of German EV registrations and fast charging stations.

On the supply side, CPOs set network-wide prices and plan fees to maximize static profits under Bertrand-Nash competition, taking the location of their stations as given. I estimate the demand and supply parameters jointly by GMM. The moments combine three sources of information. The first is station-level demand, conditional on network fixed effects and instrumented with the number and quality of competing stations nearby. The second is micro moments from the survey: public charging shares by access to home and workplace charging, work charging shares by the price difference between work and home charging, and plan shares by provider. The third is the first-order condition of the CPOs, evaluated at a marginal cost equal to the commercial price of electricity.

Two parameters are central to the analysis: the disutility of detours and the price coefficient. I identify the detour coefficient following the logic of Berry, Levinsohn, and Pakes 1995 and Houde 2012. The differentiation instruments measure the number and quality of competing stations within rings of 100 meters to 5 kilometers around each station. These instruments restrict how demand at each station varies with the proximity of stations from competing networks. The rate at which demand falls with that proximity depends on how much consumers dislike detours. The price coefficient is harder to identify because prices do not vary over time or across the stations of a network. I identify it primarily from the optimality conditions of the firms. I assume that the marginal cost of each network equals the commercial price of electricity that corresponds to its volume of demand, net of the value of greenhouse gas quota credits. I assume networks operated by utility companies pay the wholesale cost of electricity, net of the greenhouse gas quota credits. For networks that set a single linear price, the moments require the marginal cost implied by the first-order condition to match this price. For the subscription networks, the moments require the first-order conditions for the basic price, the plan price, and the plan fee to hold at this marginal cost. The price coefficient must therefore rationalize both the linear prices and the subscription price discounts observed across all networks. The share of charging done at work, as a function of the price difference between home and work charging, provides a check on this value that does not rely on the supply side.

The model fits the data well. It matches the shares of fast, home, work, and public slow charging for each type of consumers. It also reproduces moments that are not targeted in estimation: within each type, the share of public fast charging increases with annual mileage, broadly matching the patterns in the survey. The average station-level price elasticity is -1.79, close to the experimental estimates of Bernard et al. 2025. Drivers without home or workplace charging are the least elastic. The demand estimates imply that subscriptions lock in heavy users. Plan subscribers use fast charging 2.8 times more than non-subscribers, and a subscriber does 26.4% more charging at the network of their plan than they would without a plan. Plan subscribers are far less elastic at in-network stations, and diversion ratios to in-network stations are far higher for subscribers compared with non-subscribers. Banning nonlinear pricing lowers the sales-weighted average price by 9% and raises the quantity of fast charging by 5.5%. Non-plan networks cut their price by more than plan networks (-10.7% versus -5.9%), which is consistent with subscriptions softening the response of rival networks. Profits fall by 18.2% for plan networks and by 3.2% for non-plan networks. Most consumers gain from the ban, including most subscribers, because all networks lower their linear prices.

Moreover, I conduct another set of counterfactuals which quantify the complementarities between network size and the ability to price discriminate. For each plan-offering network, I reallocate a fraction (ranging between 10 and 75%) of its stations at random to its competitors. This will decrease networks' density and cause their route coverage to deteriorate. I then solve for the new equilibrium, both with and without subscription pricing. As expected, profits fall as networks become smaller and their route coverage degrades. The gain from offering subscription pricing increases with network size and is concave, suggesting diminishing returns to network size. These results illustrate the complementarities between network differentiation and price discrimination. Larger, more attractive networks are able to utilize price discrimination more effectively than smaller networks. Price discrimination using subscriptions plans enables dominant networks to capitalize on advantageous positions.

1.1 Related Literature

One major contribution of this paper is to study price discrimination in two-part tariffs within an asymmetric oligopoly setting. There is an extensive literature on price discrimination. Theoretical contributions about the welfare implications of third-degree price discrimination in a monopoly setting include Robinson et al. 1969, Varian 1985, and Aguirre, Cowan, and Vickers 2010. A comprehensive survey can be found in Lars A Stole 2007. In an oligopoly setting, relevant papers include Holmes 1989, and Corts 1998. Empirical papers examining third degree price discrimination in oligopoly settings include Borenstein and Rose 1994, Asplund, Eriksson, and Strand 2008, Gerardi and Shapiro 2009.

While third-degree price discrimination has been studied extensively, there has been far less analysis of competition in two-part tariffs in an oligopoly setting. Competition using two-part tariffs in oligopoly has been studied in a theoretical context by Armstrong and Vickers 2001, Rochet and Lars A. Stole 2002, Yin 2004, Griva and Vettas 2015, and Tamayo and Tan 2021. On the empirical side, relevant papers include Miravete and Röller 2004, Miravete 2005 (monopoly) and Davies, Price, and Wilson 2014 (oligopoly).

My setting is unique in a few respects. I provide an empirical evaluation of nonlinear pricing in oligopoly where preferences across firms are correlated with consumers' types. Theoretical papers studying such a setting include Lars A. Stole 1995 and Gomes, Lozachmeur, and Maestri 2026. Another key difference in my setting is that not all firms use subscriptions: only the major providers do. This ties to the theoretical settings of Champsaur and Rochet 1989, Calzolari and Denicolò 2015, and Chao, Tan, and Wong 2022, who study nonlinear pricing when firms are vertically asymmetric and therefore differ in their ability to price discriminate. Consistent with their theoretical findings, I show that nonlinear pricing under asymmetric competition can have anticompetitive effects.

I also contribute to the empirical literature involving competition and nonlinear pricing. Relevant studies in this context include Busse and Rysman 2005 (Yellow Pages advertising), McManus 2007 (specialty coffee), and Crawford, Shcherbakov, and Shum 2019 (cable television). Studies of loyalty programs and price discrimination in airline markets include Berry, Carnall, and Spiller 1996, Lederman 2007, and Lederman 2008.

My paper also contributes to the growing literature related to electric vehicles and charging infrastructure. This includes a number of works which have modeled the complementarity between charging infrastructure and EV adoption (S. Li et al. 2017, Springel 2021, J. Li 2023, Remmy 2026), consumers' demand for EV charging stations (Dorsey, Langer, and McRae 2025, Barwick, S. Li, and Xia 2026), and the price sensitivity of charging demand (Bailey et al. 2023, Metcalfe et al. 2026, Heid, Remmy, and Reynaert 2025, Bernard et al. 2025).

Most papers discussing the price sensitivity of EV charging demand focus on the temporal aspect of demand, highlighting consumers' responsiveness to time of use pricing and their willingness to shift charging to off-peak times. In my setting, this is less relevant: the price of fast charging almost never varies by time of day, with networks offering fixed prices per kWh. Additionally, while Bernard et al. 2025 estimates price sensitivity for public charging using a large-scale randomized control trial, they do not model consumers' station choice.

My paper contributes to this literature by estimating demand for EV fast charging using a novel dataset of real-time charging data for virtually all fast chargers in Germany. Given the distinct lack of publicly available fast charging demand data, there has been little empirical work examining fast charging. By estimating key demand charging demand parameters, my paper contributes to the understanding of demand for fast charging and how it varies by consumer type. Understanding how consumers with different available charging options make charging decisions is critical for informing policies targeting EV adoption.

2 Background and Data

2.1 Where do consumers charge?

Just like internal combustion engine (ICE) vehicles need fuel, EVs need electricity. While ICE vehicles can only get fuel from a gas station, EV drivers have a variety of options for where to recharge their vehicle. Charging can occur either in private (at home or the workplace) or in public (at either slow or fast charging stations). Consumers differ in the availability of these options.

The majority of EV charging occurs either at home or the workplace. Drivers with home charging simply plug their vehicle into a wall socket or charging point and wait for it to recharge. It is common to plug in one's vehicle at night and wake up in the morning with a full battery. Other drivers can recharge at the workplace. These drivers need only plug in at the beginning of the workday and drive home with a full battery. Consumers with either of these options rely primarily on these options due to their convenience. However, not all consumers have access to these options. For instance, those living in an apartment may not be able to charge at home. Similarly, employers differ in their provision of charging infrastructure.

EV drivers may also charge their vehicle while they are on the go using a public charging station. Public charging stations can be found in a variety of locations including parking lots, stores, restaurants, and near the highway. Public charging stations are divided into two groups: slow (AC) and fast (DC). The primary tradeoff is the speed of charging: AC charging takes hours to charge a vehicle, while DC chargers can recharge the majority of a battery in 15 to 30 minutes. AC charging is suitable for situations where the consumer expects to park for an extended period and is reasonably close to their destination. In contrast, DC charging allows consumers to charge much faster and facilitates charging en route. Fast charging stations are commonly found along highways and primary roads, but they may also be found near restaurants, grocery stores, and other business establishments.

This paper focuses on fast charging. A fast charging station is similar to a gas station: it has a collection of chargers, just like a gas station has multiple pumps. This enables multiple consumers to use a particular station at the same time. Fast charging stations are owned by Charge Point Operators (CPOs). CPOs operate a number of charging stations referred to as a *network*, similar to how a gas station brand operates multiple locations. I will use the term network and CPO interchangeably throughout this paper with the understanding that a CPO refers to the operator of a charging network.

The 'speed' or 'power' of a fast charging station is described in terms of kilowatts (kW). A kilowatt is a measure of the rate of electricity usage or delivery. A kilowatt-hour (kWh) corresponds to the quantity of electricity used or transmitted by one kilowatt of power over the course of an hour. In other words, a kilowatt is a measure of flow while a kilowatt-hour is a measure of stock. Charging speeds are represented in terms of kilowatts, while charging amounts and battery sizes are discussed in terms of kilowatt-hours.

Fast chargers vary in their maximum rate of power delivery. Available speeds range from 50 kW to upwards of 350 kW. Chargers with higher speed offer the potential for significantly faster charging. However, the realized speed also depends on the maximum rate of power the consumer's vehicle can accept. Cars vary significantly in their maximum acceptance rate, which will limit their realized charging speed.

2.2 Consumer heterogeneity: USCALE survey

A central feature of the EV charging market is the large degree of consumer heterogeneity. Some drivers can charge at home or work, while others cannot. Some are highway drivers while others are city drivers. Some drive 5,000 km, whereas others drive upwards of 25,000 km. Some can charge faster, while others charge much slower. All of these factors influence 1) how frequently consumers need to charge, 2) how often they charge in public, and 3) which chargers they choose to charge at.

I develop a model which incorporates each of these dimensions of heterogeneity. This model relies heavily on data which can speak to the rich variation in consumers' types and charging behavior. To address this, I use detailed survey data from USCALE, a German company which specializes in surveys regarding electromobility.[2] Specifically, I use USCALE's 2024 Public Charging Study as my main data source to model the distribution of consumer types. The Public Charging Study is a survey of 2,986 German EV drivers. The survey was conducted online from July-September 2024. I impose two main restrictions on the survey data I use for sample construction and estimation. First, I exclude respondents driving a company vehicle, as my focus is on private consumers. Second, I drop Tesla drivers from the sample. This is because Tesla drivers do most of their public charging at the Tesla Supercharger network, which I am not modeling in this paper. After making these restrictions, I end up with a sample of 1,878 drivers.

As a supplement, I also use the 2024 eMSP Loyalty Benchmarking Study, which was conducted from August to September 2024 and features 2,688 drivers. I apply the same sample restrictions (private vehicles, no Tesla drivers) and end up with 1,822 drivers. This data is not available in a tabular data format but is rather available as an online dashboard that presents a collection of summary statistics. This data reports the share of consumers using a paid plan when charging at a given network. It also reports drivers' preferred charging provider.

The Public Charging Study contains a wealth of information about consumers and their vehicle. The survey reports coarse levels of consumer demographics, including their federal state, income bin, age, residence type (single vs. multifamily home, rent vs. own, parking availability), and city type. I also see key information about consumers' vehicle. Namely, I observe the EV make and model, EV range (binned), battery size (binned), and maximum charging speed (binned). Crucially, the survey also contains information on consumers' driving and charging behavior. For driving behavior, I see binned average daily commute distance and binned annual mileage. The data also lists which charging options are available to a consumer (home, work, and public) as well as the share of charging they do at each option. This allows me to quantify how many people have access to charging at home and/or work. It also shows to what extent they use either option when they are available. Finally, I observe the price difference between home and work charging, when both are available. These variables will be critical for modeling heterogeneity in consumers' charging behavior.

Using this information, I construct consumer groups based on the charging options available to each consumer. Consumers of type h have access to charging at home, but not at work. Similarly, consumers of type w have charging at work but not at home. Consumers of type hw have access to charging at both home and work, while consumers of type n have access to neither.

Table 1 presents a collection of summary statistics for each of these four consumer groups. Each column corresponds to consumers with different available outside options (hw, h, w, and n). The first group of statistics reports the share of drivers in each consumer group who live in each city type. For clarity, I group these into large, medium, and small based on

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[2] https://uscale.digital

	Outside option group			
	hw	h	w	n
City type				
Large	0.46	0.26	0.57	0.54
Medium/Suburb	0.21	0.24	0.19	0.25
Small/Countryside	0.33	0.51	0.24	0.20
Annual mileage				
$\leq$ 10,000 km	0.10	0.28	0.22	0.35
10-20,000 km	0.51	0.52	0.50	0.49
$>$ 20,000 km	0.39	0.20	0.29	0.16
Charging shares				
Home	0.53	0.84	-	-
Work	0.31	-	0.60	-
Public fast	0.12	0.13	0.25	0.52
Public slow	0.05	0.03	0.15	0.48
N	421	1142	129	186

Table 1: Summary statistics for each consumer outside option group. The first group of statistics presents the share of consumers in each city size bin for each outside option group. The second group of statistics represents the share of consumers within each annual mileage bin by outside option group. The final group of statistics lists the average share of energy obtained from each charging source by outside option group.

the types reported in the survey.[3] The main takeaway is that consumers without access to charging at home tend to reside in larger cities: the majority of w and n consumers live in large cities. Consumers with home charging (h or hw), conversely, are less likely to live in large cities. Table 12 in the appendix breaks home and work charging availability by city type. The proportion of respondents with home charging decreases with the level of urbanization. Conversely, the proportion of respondents with work charging is increasing in urbanization: only 19 percent in the countryside have it, compared with 47 percent in the largest cities.

I now move on to mileage. h and n consumers are more likely to have annual mileage less than 10,000 km and less likely to drive more than 20,000 km. For h consumers, this is likely due to their tendency to live in small cities. For n consumers, this is likely because they live in large cities and commute shorter distances. It could also be due to selection: consumers with long commutes who don't have easy charging access may not select an EV. In contrast, consumers with charging at work (either w or hw) have higher mileage than h or n. This may be a selection story. Higher mileage drivers who can't charge at work may be less likely to purchase an EV, leading to a selected sample of higher mileage drivers who utilize work charging. Or, workplaces with a larger share of commuting employees may be more likely to offer charging at work.

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[3] The survey contains the following city classifications: Large city (>1M), Large city (> 500K), Large city (>100K), Medium city, Suburbs, Small city, and Countryside. For clarity, I group large cities together, medium city with suburbs, and small city with countryside.

Finally, I find that consumers' available options influence their charging behavior. Consumers with home charging only (h) do 84 percent of their charging at home, while consumers with charging at home and work do 53 percent at home and 31 percent at work. Consumers with only workplace charging do 60 percent of their charging there. These results are consistent with the conventional wisdom that most EV charging occurs either at home or at the workplace, with most drivers seldom relying on public charging infrastructure.

Consumers prefer to charge at home over charging at the workplace. This is evident by the larger share of charging done at home (53%) versus at work (31%) for hw drivers. This can also be gleaned by comparing the home charging shares for h consumers (84%) to the work charging shares for w consumers (60%). Why is this the case? It is likely a composition of two effects. First, work charging is only feasible on commuting trips and does not cover other trips a driver might take. This limits consumers' utilization of work charging. Second, home charging may be more desirable for a variety of reasons (convenience, price, availability). For instance, work chargers are a shared resource and may not be accessible if other employees are using them. Or, they may be more expensive or less convenient to access. Whatever the reason, consumers tend to prefer home charging when it is available.

The other major takeaway is that consumers with access to charging at home or work rely less on fast charging infrastructure. The fast charging share is 12 percent for hw consumers and 13 percent for h consumers, while it is 25 percent for w consumers. Consumers with neither home or work charging do 52 percent of their charging at fast chargers. The preference for alternative charging options is intuitive: charging at home or work is more convenient. Home and work charging are passive, since the consumer does not have to drive anywhere and does not incur waiting costs. Charging at a public fast charger requires driving to the station, waiting at the station while the car charges, and pays a higher price than they would at home or work.

Finally, I observe limited information about consumers' plan membership. In the eMSP Loyalty Benchmarking Survey, I observe aggregated statistics about consumers' preferred EV charging provider as well as the share of consumers who use a paid plan from a given provider. I will use this information to help estimate the model. Specifically, I will use the survey-implied paid plan shares to construct moments based on the share of consumers choosing one of a given CPO's paid plans.

2.3 Commuting flow and mobility data

Driving behavior is a key determinant of public charging demand. Consumers differ both in how much they drive as well as where they drive. Both of these shape the charging choices they make. Drivers with higher mileage require more charging than lower mileage drivers. The location of consumers' driving is also vitally important: consumers will choose between charging stations that are close to routes they usually drive. Two drivers driving in different parts of the same city will encounter different sets of chargers. Similarly, a highway driver has a different set of choices than a city driver.

In my model, I assume each consumer makes a set of regular trips. These routes define their charging station choice set and determine the quantity of charging they need. I assume consumers take both commuting trips as well as non-commuting trips. To accurately model which routes drivers take, I utilize commuting flow data and large-scale mobility surveys provided by the German government.

The empirical distribution of commuting trips is computed using commuting flow data for the year 2024 provided by the German Federal Employment Agency.[4] The dataset contains Origin Destination (OD) tables, listing the number of commuters between each pair of municipalities in the country. The data is derived from administrative records for all employees subject to social insurance contributions (akin to FICA taxes in the United States) and represents the vast majority of the working population. I have plotted a map of the observed OD commuting flows in Figure 1. I include all commuting trips which are shorter than 100km. One important limitation of this data is that it does not distinguish by travel mode.

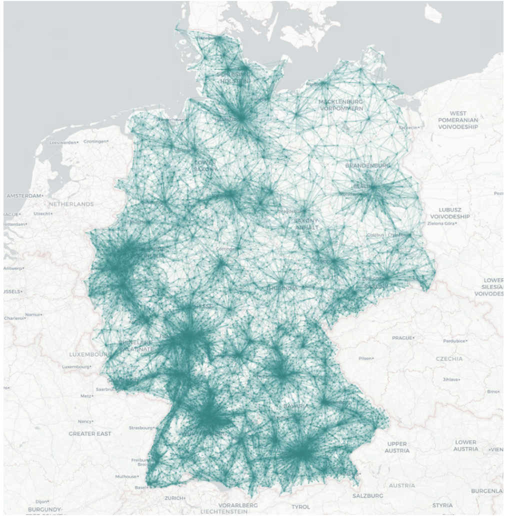
SCHLESWIG-HOLSTEIN
MECKLENBURG-VORPOMMERN
WEST POMERANIAN VOIVODESHIP
Szczecin
Bremerhaven
Hamburg
Leeuwarden
Groningen
LOWER SAXONY
BRANDENBURG
Berlin
Potsdam
AMSTERDAM
NETHERLANDS
Utrecht
Rotterdam
Münster
Magdeburg
SAXONY-ANHALT
LUBUSZ VOIVODESHIP
Zielona Góra
Cottbus - Chóśebuz
NORTH RHINE-WESTPHALIA
Göttingen
Antwerp
THURINGIA
Dresden
SAXONY
LOWER SILESIA VOIVODESHIP
HESSE
Namur
Charleroi
RHINELAND-PALATINATE
PRAGUE
Pardubice
LUXEMBOURG
Luxembourg
Pilsen
CZECHIA
Jihlava
Saarbrücken
Metz
Brno
BAVARIA
Nancy
Strasbourg
GREATER EAST
UPPER AUSTRIA
LOWER AUSTRIA
Freiburg im Breisgau
Mulhouse
Basel
AUSTRIA
ZURICH
Dijon
BURGUNDY
VORARLBERG
LIECHTENSTEIN
TYROL
SALZBURG
STYRIA

Figure 1: Sampled commuting flows, aggregated by origin and destination municipality code.

I account for non-commuting trips using the Mobility in Germany 2023 survey (BMV 2025) conducted by the German government. The survey is a large-scale, cross-sectional travel diary survey with responses from over 420 thousand individuals in over 172 thousand households. Each participating household is assigned a reference date on which they are to record the full set of trips taken for each individual in the household. I observe around 1 million trips in total. This dataset reports the type of trip (commuting, leisure, shopping, etc.), travel mode, distance, and travel time, as well as coarse information about the household. Importantly, I can observe the type of city the household resides in. This allows me to model the distribution of trip distance conditional on region type.

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[4] https://statistik.arbeitsagentur.de/DE/Navigation/Statistiken/Interaktive-Statistiken/Pendler/Pendler-Nav.html

I sample both commuting and non-commuting trips from their empirical distributions. For commuting trips, I use the commuting flow data in conjunction with the mobility data to back out the distribution of car commuting trips. For non-commuting trips, I use the mobility data to sample four non-commuting trips for each consumer. The appendix contains further details on how I sample each type of trip from its empirical distribution. This procedure yields a set of origin and destination coordinates corresponding to one commute and four non-commute trips for each consumer.

Given a sampled origin and destination location (for either commuting or non-commuting trips), I assume drivers take the route through the road network which minimizes their travel time. I utilize the implementation of Dijkstra's algorithm in the `osmnx` Python package (Boeing 2025), which accounts for the driving speed on different road types. I abstract away from considering the effect of traffic or idiosyncratic route preferences. Information on the German road network is obtained from OpenStreetMap (OSM). OSM is an open-source mapping project which freely offers comprehensive information on road networks, amenities, and points of interest (OpenStreetMap contributors 2024). For each of these constructed routes, I find all fast chargers within a 5 km buffer. This represents the choice set of fast chargers for this route. For each of these stations, I compute the detour time needed to access that station from the original route, assuming the consumer takes the optimal detour. This procedure yields route-specific charger detour times for each charger within 5 km of the route. This process is repeated for each of the routes the consumer takes. These detour times are used in the estimation of my model.

2.4 Consumer locations and vehicle characteristics

I use vehicle registration data from the German Federal Motor Transport Authority (KBA) to both measure how many EVs are on the road in a given area and uncover the distribution of vehicle characteristics. The agency publishes a variety of statistics regarding the existing vehicle fleet as well as new vehicle registrations. I employ two of these reports for my analysis.

First, I use the KBA's FZ 27 dataset, which reports the stock of electric vehicles within each zipcode on a quarterly basis. This dataset includes every vehicle on the road, not just newly registered vehicles. This allows me to model the spatial distribution of EV drivers at a very granular level. I use the dataset to determine how many EV drivers reside in each zipcode, which are used construct representative weights for the simulated consumers in my model.

In addition, the FZ 6 dataset reports the aggregate stock of each make, model, and trim level on an annual basis. This data is aggregated to the country level and therefore does not give information on local market shares of each vehicle. I assume national vehicle market shares are representative of local shares. I match this dataset with information on each EV model's battery size, effective range, and maximum charging speed scraped from chargeprice.app.

This data is important for my model since vehicle characteristics influence drivers' charging behavior. An EV's range determines how frequently its driver must charge. In my model, a consumer's mileage and their EV's range jointly determine the number of charging events they must complete. Another important characteristic is the maximum charging speed of the vehicle. Every EV has a maximum rate of charging, measured in kW. A vehicle cannot charge faster than this limit, even if it is plugged into a charger which can deliver power at a faster rate. The maximum charging rate of a vehicle can limit its ability to benefit from faster charging. Variation in vehicle charging speeds will affect consumers' choice of charging station and willingness to pay for faster charging speeds.

2.5 Market definition and sample period

The overlapping nature of commuting patterns makes market definition difficult. No matter what boundary one specifies to demark a market, there will always be some consumers from beyond that region who commute into it. For instance, a market boundary intersecting a major highway separates chargers into two separate markets. However, since some consumers pass by chargers on both sides, those chargers which shares consumers in common should be considered to compete within the same market. Even if one distinguishes between city or highway chargers and classifies markets in this way, drivers commuting to and from the city (or between two nearby major cities) would potentially use chargers from both the highway and city markets.

This problem is inescapable, but I aim to minimize it by considering relatively broad markets. When constructing markets, I first use the EU's Functional Urban Area classification[5]. Functional Urban Areas (FUAs) are defined as cities and their associated commuting zone. In certain regions (e.g. the Rhine-Ruhr area), there are multiple FUAs located right by each other. As can be seen in Figure 1, there are many overlapping commuting flows between major cities in the same region. Therefore, I decide to group adjacent FUAs together and define the convex hull of their union as a 'market'. I consider all chargers within this region as belonging to the same market and compute market shares accordingly. This approach minimizes the market truncation issue.

For my analysis, I use the Rhine-Ruhr Metropolitan Region, Hessen and Baden-Württemberg, the Hamburg Metropolitan Region, the Central German Metropolitan Region, and the Munich Metropolitan Region. These regions together account for around 54 percent of total German EV registrations and 48 percent of fast charging stations. A map of these regions can be found in Figure 11. These markets are chosen because they represent major commuting markets and are distinct from each other and their surrounding regions. They also contain both dense, urban areas as well as smaller cities. I use the sample period of September 2024 for my analysis. I select this period to align with the conclusion of the USCALE consumer survey.

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[5] https://ec.europa.eu/eurostat/statistics-explained/SEPDF/cache/72650.pdf

2.6 Charging networks and demand

I construct a novel dataset of real-time charging station utilization for the vast majority of German EV fast charging stations. This dataset includes all stations which publish real-time information through the Open Charge Point Interface (OCPI). My data ranges from November 2023 until November 2025. For the purposes of this paper, I will be analyzing demand from the month of September 2024. Notably, this data does not include the Tesla Supercharger network. The Supercharger network is available to all EV drivers but unfortunately does not make data available via the OCPI. This makes scraping real time information significantly more challenging. I thus omit the Supercharger network from my analysis.

I scraped the data in real time at approximately three-minute intervals. In each round of scraping, I observe the number of available charging points at a given station. As I observe the number of available charging points over time, I gain a high-frequency profile of charging behavior at the station and can back out the number of charging events that occur. I define a charging event as a decrease in the number of available charging points within a station between two consecutive time periods.

Charging demand — I now provide a summary of the major players in the German fast charging market. Table 2 lists summary statistics for the top 10 networks in my sample regions as of September 2024. It reports the number of stations in each network, the number of observed charging events, and the number of charging events per station. The table is sorted by the observed number of charging events in decreasing order. CPOs can generally be classified into four major categories: public utilities and power providers (EnBW, EWE, E.ON), oil and gas majors (Aral Pulse and Shell), grocery (Lidl, Kaufland, ALDI Süd), and pure-play providers (Ionity and Allego).

Two things are worth noting in this table. The first is the asymmetry in network size. The largest, EnBW, is a large utility company which operates a nation-wide network of stations. The smallest in this table is Ionity, a joint venture of several automakers, with 61 stations. While Ionity is the smallest of the top 10 in terms of station count, its locations are almost entirely located along the Autobahn and other major highways. The second main takeaway is the vast difference in utilization (charging events per station) between the top three networks (EnBW, Aral Pulse, and Ionity) compared with the other top networks. Each of these networks achieves significantly higher utilization than other networks.

Table 3 reports summary statistics of station characteristics for each network. It includes the average number of charging ports per station, the average maximum power (measured in kW), the share of stations within 1 km of the Autobahn, the share of stations at an Autobahn service area, and the share of stations at Autobahn truck stops. Networks vary in the average

Network	Stations	Charge events	Charges/station
EnBW	686	290,808	423.92
Aral Pulse	241	125,590	521.12
Ionity	61	67,023	1,098.74
Shell	228	63,948	280.47
Lidl	269	63,137	234.71
Kaufland	184	55,321	300.66
EWE	271	55,211	203.73
ALDI	180	48,179	267.66
E.ON	174	36,659	210.68
Allego	195	33,603	172.32

Table 2: Network size and monthly aggregate demand for the top 10 networks in my sample markets. The time period is September 2024. The table is sorted by the number of observed charging events in decreasing order.

Network	Ports/station	Power (kW)	Autobahn (1km)	Service area	Truck stop
EnBW	5.50	172.57	0.37	0.07	0.02
Aral Pulse	7.06	211.99	0.46	0.02	0.09
Ionity	7.52	351.37	0.95	0.64	0.11
Shell	6.43	241.92	0.35	0.01	0.03
Lidl	3.19	45.06	0.25	0.00	0.00
Kaufland	3.49	43.85	0.22	0.00	0.00
EWE	2.38	207.31	0.39	0.00	0.04
ALDI	2.82	94.20	0.42	0.00	0.00
E.ON	4.58	97.29	0.73	0.55	0.01
Allego	4.12	144.44	0.51	0.00	0.08

Table 3: Summary statistics of network characteristics. For each network, I report the average number of charging ports per station, the average maximum power (measured in kW), the share of stations within 1 km of the Autobahn, the share of stations at an Autobahn service area, and the share of stations at Autobahn truck stops.

size of station: EnBW, Aral, Ionity, and Shell have the largest stations, whereas grocery-backed networks Lidl, Kaufland, and Aldi tend to have smaller stations. Networks also differ in their average maximum charging power. There is essentially a bimodal distribution, with EnBW, Aral, Ionity, Shell, EWE, and Allego offering faster charging while other networks (Lidl, Kaufland, Aldi, and E.ON) offering much slower charging on average.

Networks also vary in their presence along the Autobahn highway network. Most of the non-grocery networks having at least a third of their stations within 1km of the Autobahn. Nearly half of Aral's stations are Autobahn-adjacent, while 95 percent of Ionity's are. While Ionity has comparatively fewer stations, they are more strategic in their site choices and tend to target higher-traffic locations along major highways. This strategy targets consumers who are driving longer distances between major cities rather than those driving within a given city.

In addition, a select few networks have a presence at Autobahn service areas. Service areas are located directly on the Autobahn and offer many amenities including fast food, restrooms, and parking. The vast majority of Autobahn service areas are owned and operated by Tank & Rast. Tank & Rast partnered with CPOs EnBW, Ionity, E.ON, and Mer to offer fast charging at their service areas. Other networks were excluded from service areas, with a few exceptions: gas-affiliated networks were able to build chargers on gas station forecourts, but could not build standalone stations.[6] As a result, none of the other networks have any presence at Autobahn service areas. This is consequential, since chargers at service areas get significantly higher utilization.

CPOs also vary in their presence at Autobahn truck stops (Autohofs). These are privately-owned rest areas located directly off of highway exits. Like service areas, they feature ample amenities and are convenient for drivers. While the name implies that they are suited for semi-truck drivers, they are commonly used by motorists as well. Aral operates a number of truck stops, giving them a natural location to construct charging stations. Other networks such as EnBW, Ionity, Shell, EWE, E.ON, and Allego have a presence at truck stops, to varying degrees.

Concentration — The market for charging is concentrated. While national market shares are not overly concentrated, the story is quite different at the local level. The average city in my sample has 3.81 stations and 2.49 networks. The average city-level Hirschman-Herfindahl Index (HHI) is 7,486.23, while the median is 10,000 (only one network). This suggests very little competition and high concentration at the city level. Granted, the city level is not necessarily the relevant market definition. Since many consumers' routes cross city boundaries, city-level metrics will understate the number of choices consumers face.

Radius (km)	# stations	# networks	Mean HHI	Median HHI	SD HHI
1	2.39	2.07	7,347.58	7,474.53	2,571.42
5	13.87	7.27	3,664.08	2,881.02	2,279.02
10	37.49	12.81	2,444.99	1,909.18	1,581.28
20	100.44	20.96	1,619.65	1,389.35	778.75
City	3.81	2.49	7,486.23	10,000.00	2,892.76

Table 4: Summary statistics of concentration metrics for catchment areas of varying sizes (1, 5, 10, and 20 km). Catchment areas of varying sizes are defined for each station. The mean number of stations and networks are recorded. Market shares of each network are computed within the catchment area by aggregating each network's demand, which are then used to compute the HHI at the catchment level. The mean, median, and standard deviation of catchment-level HHI are presented in the final three columns.

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[6] This exclusion sparked a 2022 lawsuit that turned into a lengthy legal battle in the German courts. It was ruled in 2026 that this allocation of stations was illegal and that future stations slots must be auctioned off competitively

To provide a more sensible measure of local concentration, I calculate station-level statistics using catchment areas of varying radii around each station. For each station, I aggregate demand at the network level for all stations within 1, 5, 10, and 20 kilometers. I then compute market shares and HHI for each catchment area. This approach accounts for the presence of local competition at various levels and abstracts away from geographic boundaries.

On average, there are 2.39 other stations and 2.07 networks within 1 km of each station, indicating that most stations face very few nearby competitors. Within 5 km, there are an average of 13.87 stations and 7.27 networks. As the radius expands, the numbers of stations and networks both increase. Within 20 km, there are an average of 100.44 stations and 20.96 networks.

The average HHI within 1 km of each station is 7,347, indicating significant concentration. As the radius expands, the HHI decreases from an average 3,664.08 at a radius of 5 km down to 1,619.65 for a radius of 20 km. There is also significant dispersion in HHI: the standard deviation decreases from 2,279.02 for a radius of 5 km to 778.75 for a radius of 20 km. The market is highly concentrated even when considering all stations within 10 km of a given station, and is moderately concentrated when the radius doubles to 20 km. Even when there are many alternative networks, the market is highly concentrated. It is worth noting that the average consumer is not likely to consider every station within 20 km. Rather, they will consider the stations close to the route that they are taking.

Given that local markets are so concentrated, the natural question is: why? More specifically, why do certain networks capture an outsized share of demand compared to others? Part of this is due to differentiation: some stations are more conveniently located than others or may offer faster charging than nearby competitors. For instance, a station located right off the highway will expect to attract more demand than one that requires a 5 minute detour. Since networks differ in the types of locations they enter, this creates asymmetries in demand. Similarly, faster charging networks can expect to be favored by consumers with faster charging vehicles. When faced with a 350 kW charger and a 50 kW charger, a consumer who can make use of the faster charger will save a great deal of time if they pick the faster one. Both sources of differentiation can explain why some networks attract more demand.

However, I will argue that network differentiation alone is insufficient to explain the observed concentration. Figure 2 plots networks' average share of demand versus their share of stations within a 10 km radius, averaged across stations within the network. I plot EnBW, Ionity, and Aral Pulse on their own, and I plot the average over all other networks in grey. A 45-degree line is included for comparison. These three networks - each of which offers subscription plans - attract a far larger share of charging demand than their station share would suggest. For instance, when they have 25 percent of stations within a 10 km radius, they receive anywhere between 50 percent (Aral Pulse) and 65 percent (Ionity) of demand in that region. Consumers systematically favor these networks over competitors. The previous section discusses how these networks are differentiated from their competitors and offer fast

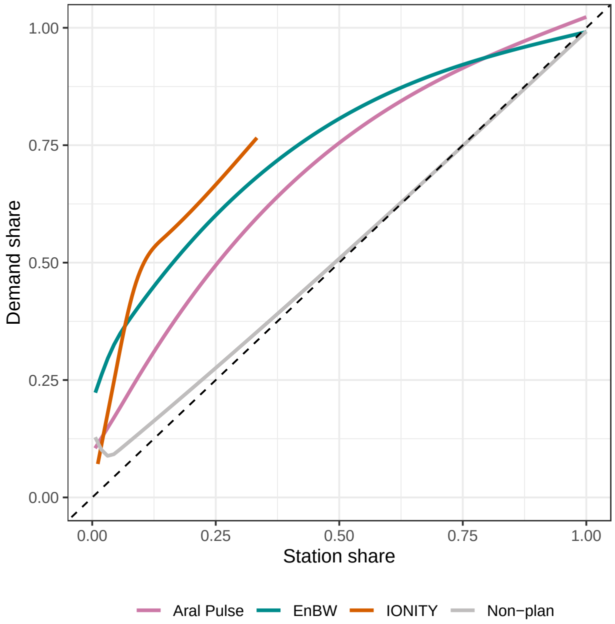

Figure 2: Average market share of each network plotted against the network's share of stations within 10 km. EnBW, Ionity, and Aral Pulse are plotted separately, while non-plan networks are aggregated. A 45-degree line is plotted for reference. Each line represents the conditional mean computed using LOESS.

charging at convenient locations. However, the usage of subscription pricing by these three networks is also contributing to their local dominance. Subscription plans incentivize heavy charging consumers to do more charging in-network, which boosts utilization.

Pricing — I now discuss the structure of charging prices. I obtained pricing data from the website chargeprice.app. Prices for public charging are set by each charging network. Virtually all networks at the time of my sample set uniform prices across the entire country - that is, almost all networks have no spatial (or temporal) variation in prices. This is in contrast both with the market for gasoline (where prices are spatially differentiated) as well as the market for electricity (where prices vary based on time of use). Some networks set prices on a country-by-country basis, but very few set prices on a more granular level.

Most CPOs simply set one price, which I will refer to as the *basic price*. That is, a per-kWh charging price available to all consumers without any monthly subscription. This price can be obtained usually by paying through the provider's own app (or card). Often, the consumer must set up a free account with the provider as well. As an example, consumers can charge at an EnBW station for €0.59/kWh so long as they download the EnBW mobility+ app and create a free account. I assume that the consumer pays the cheapest available price from a given provider. I rule out ad-hoc charging, which allows charging without an account

Network	€/kWh
EnBW	0.59
Aral Pulse	0.61
Ionity	0.69
Shell Recharge	0.61[8]
Lidl	0.44
Kaufland	0.47
EWE	0.59
ALDI Süd	0.39
E.ON	0.61
Allego	0.73

(a) Per-kWh non-subscriber prices for the top 10 networks in my sample markets as of September 2024, sorted in decreasing order by charging demand.

	€/kWh	Monthly fee (€)
EnBW	0.59	0
	0.49	5.99
	0.39	17.99
Ionity	0.69	0
	0.49	5.99
	0.39	11.99
Aral	0.61	0
	0.57	4.5[9]

(b) Per-kWh prices and monthly plan fees for major subscription providers. All values are in Euros. The non-subscription prices for each network are included for comparison.

Figure 3: Per-kWh non-subscriber prices and plan-specific prices and fees.

but is often significantly more expensive.[7] I also exclude roaming plan prices, which offer consumers the ability to pay for charging at a variety of networks. Roaming prices are often flat across networks (i.e., they can charge at any network for €0.75/kWh) or group networks into different pricing tiers (low price, high price). Roaming prices tend to be higher than the price available in a network's app.

The basic prices for the top 10 networks in my sample are included in Table 3a. Among the top networks, the price ranges from €0.39/kWh (Aldi Süd) to €0.73/kWh (Allego). The average price for all networks in my sample is €0.61/kWh. Most networks set prices in a relatively narrow range: the first quantile is €0.58/kWh, while the third quantile is €0.65/kWh. To contextualize these prices, the average household electricity rate in my sample is €0.39/kWh, making fast charging a more expensive option than charging at home. The lowest-priced providers are grocery-affiliated networks (Lidl, Kaufland, and Aldi Süd). Their aggressive pricing is likely a loss-leader strategy to induce drivers to shop at their stores. While the vast majority of charging networks utilize uniform spatial pricing, one exception is the Shell Recharge network. During the sample period, prices at Shell Recharge stations ranged between €0.57-0.64/kWh, depending on the station's location. As I don't observe the effective station-specific price, I take a simple average price of €0.61/kWh to capture the midpoint of the price range.

Some networks - EnBW, Aral, and Ionity - offer optional subscription plans in addition to their basic charging prices. These subscription plans give enrolled consumers lower charging prices in exchange for a fixed monthly fee. Table 3b outlines the plan structure for each of these three providers. It also lists the basic price of each network for comparison. Paid plans offer a significant reduction in charging costs: subscribing to the € 5.99/month plan for EnBW reduces per-kWh charging costs by approximately 17%, while the € 17.99/month plan reduces charging costs by 33%. The € 5.99/month plan from Ionity reduces prices by around 29%, while the € 11.99/month plan offers an approximately 43% reduction.[9] Aral also offers reduced charging costs of € 0.57/kWh through its partnership with the ADAC in exchange for an annual membership fee of € 54. To represent this in comparable terms, I prorate this to a monthly basis.

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[7] Ad-hoc prices for major networks are quite high: EnBW charged €0.87/kWh for ad-hoc compared with €0.59/kWh in their app (47% higher), and Aral charges €0.79/kWh compared with €0.61/kWh in their app (30% higher). In practice, ad-hoc charging is rarely utilized by consumers: industry anecdotes and consumer surveys support this claim.

[8] Shell set station-specific prices, with prices ranging from €0.57-€0.64/kWh depending on the station.

These subscription plans target consumers who do a lot of fast charging. A plan is only worth subscribing to if one can charge enough at the selected network to break even. A consumer who charges multiple times a month will save enough from the discounted rate to cover the upfront fee. Conversely, a consumer who only charges once a month would pay more under the plan than if they were to pay the non-subscriber rate. The plan operates as a self-selection mechanism for consumers with high expected fast charging demand.

Which kind of consumers have high expected fast charging demand? My survey data indicates that it is primarily two groups: 1) consumers without access to convenient alternative charging options and 2) high mileage consumers. These groups rely on fast charging to a significant extent and are exactly the type of consumers who would benefit from subscription plans. In contrast, drivers who have home charging are far less likely to do enough fast charging to justify paying the subscription fee, unless they do a significant amount of driving.

In my model, I will assume that each network offers one subscription plan. EnBW and Ionity both offer the € 5.99/month plans which give an in-network price of € 0.49/kWh. I assume Aral offers a € 4.50/month plan that gives an in-network price of € 0.57/kWh. I exclude the higher tiers (e.g. EnBW's € 17.99/month plan and Ionity's € 11.99/month plan).

Location characteristics and amenities — To proxy for location type and the presence of nearby amenities, I use the OSM data to find all amenities and points of interest near each charging station. The availability of services (food, restrooms, coffee, etc.) nearby a charging station may play an important role in the charging choice. I account for this possibility in my model. I also determine whether the charging station is located at an Autobahn service area or truck stop. Further details of all the variables I construct using the OSM data are included in the estimation methodology section.

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[9] The affiliated automakers which comprise Ionity also offer brand-specific subscription plans for charging at Ionity. These plans are very similar to Ionity's standalone offers. For simplicity, I omit these from the analysis.

[10] Aral discount available to ADAC consumers; cost of ADAC membership is € 54/year, corresponding to € 4.50/month.

3 Model

I model both the demand and supply side of the market for EV fast charging. On the demand side, consumers choose where to charge based on their driving patterns, available charging options, and vehicle characteristics. I assume a nested BLP-style demand model which incorporates rich consumer heterogeneity. Additionally, consumers may elect to purchase a monthly subscription from a CPO which lowers their charging costs within the CPO's network. On the supply side, CPOs set charging prices and (if applicable) plan fees to maximize profits under the assumption of Bertrand-Nash competition.

3.1 Consumers

I model consumers' demand for both public and private charging using a nested logit framework with individual heterogeneity in the style of BLP. Each consumer first chooses whether to subscribe to any of the monthly plans offered by CPOs. These plans reduce the cost of charging at the CPO in exchange for a monthly fee. The utility from each plan depends on the inclusive value of charging conditional on picking that plan as well as the plan fee. The inclusive value of charging is derived from a nested logit model of consumer charging. In this model, consumers take a variety of trips which require a certain amount of charging. Consumers then choose where to charge on each charging event to maximize their utility. They choose between slow charging outside options (home, work, and public slow charging) or public fast charging (the inside good). Their choice set of public fast chargers is determined by the set of chargers adjacent to each route.

The number of required charging events is denoted by Q_i. Q_i is determined based on the total kWh used across their routes, depending on route length and their vehicle's efficiency (kWh/km). Specifically, letting D_i indicate electricity used and B_i denote battery size, the total number of (normalized) charging events is given by $Q_i = \frac{D_i}{0.6B_i}$. 0.6 is included in the denominator to reflect the average amount of the battery charged in a fast charging session. I assume each charging event replenishes 60% of the consumer's battery, corresponding to charging from 20-80% of battery capacity. This is because consumers rarely charge fully at fast chargers; the speed of charging slows significantly after 80%.

The consumer chooses where to do each of their Q_i charging events to maximize their utility. Consumers can choose to charge at public fast chargers along the route, public slow chargers, home (if available), or at work (if available). I use a nested structure to model consumers' choice between these heterogeneous options. I include home, work, and public slow charging in one nest and include all public fast chargers in a separate nest. Then, I assume consumers make a choice of charging type (fast or slow) based on the inclusive value of each type. Finally, consumers choose a plan at the beginning of the month based on their expected utility under each plan.

In short, the consumer's choice problem takes the following sequence:

1. Consumer picks plan to maximize expected utility from charging
2. Conditional on plan membership, for each required charging event, consumer decides on charging type (slow or fast)
    - If slow:
        (a) Home, work, or public slow charging (depending on availability)
    - If fast:
        (a) Which charger to charge at

I summarize the nonlinear demand parameters in my model into the vector θ:

$$\theta = (\alpha, \alpha_{plan}, \alpha_{mileage}, \beta_{detour}, \beta_{speed}, \sigma_{fast}, \sigma_{slow}, \sigma_{type}, \sigma_{paid}, v_h, v_w, v_s)$$

I will discuss each parameter as they are introduced. There are additional linear parameters that are estimated during the demand estimation procedure.

I felt it most natural to walk through the consumer's utility maximization problem in reverse order, starting with their utility from each charging event. I then work my way backwards through the route and charging type choice. I conclude by discussing consumers' plan choice.

Fast charging utility — I begin with the choice of charger conditional on taking a given route.

Consumers are characterized by their set of driving routes R_i as well as a type vector $\omega_i = (\lambda_i, \kappa_i, q_i, Q_i, p_{ih}, p_{iw})$. λ_i represents the set of outside options which are available to the consumer. These include home (h), work (w), and public slow charging (s). κ_i refers to the maximum charging speed (kW) of consumer i's vehicle. Letting B_i denote the size of their vehicle's battery, $q_i = 0.6B_i$ represents consumer i's charging demand (in kWh) per public charging event under the assumption that consumers charge from 20-80%. Finally, Q_i reflects the number of charging events (of size q_i) needed to fulfill the drivers' charging needs from the routes they take. p_{ih} and p_{iw} refer to the home and work charging prices that the consumer faces (if available).

The price that a consumer pays for public fast charging depends on their choice of charging subscription (if any) and the network of the station they are charging at. Each consumer chooses a plan n from a set of plans $N = \{0, 1, \ldots, N_p\}$. I allow consumers to choose not to subscribe to any plan $(n = 0)$. Instead, the consumer pays each CPO's basic per-kWh price whenever they charge at any of their stations. If, conversely, a consumer selects a plan n from CPO A, they will receive a discount when charging at stations of A and pay other CPOs' linear price when charging at other CPOs. To fix ideas, a $n = 0$ consumer would pay €0.59/kWh at EnBW and €0.69 at Ionity. A subscriber to EnBW would pay €0.49 at EnBW and €0.69 at Ionity. Conversely, an Ionity subscriber pays €0.59 at EnBW and €0.49 at Ionity. I assume that consumers can only pick one paid plan.

Consider the case of consumer i who has selected plan n and who takes the set of routes R_i. Each route $r \in R_i$ is characterized by an origin and destination point. For each route $r \in R_i$, define $C(r)$ as the set of chargers within 5 kilometers of route r. Then, let $C_i = \bigcup_{r \in R_i} C(r)$ represent the choice set of chargers available to consumer i across all of their trips. C_i is the union of consumer i's choice sets across each of the routes they take. Given nonlinear parameter vector θ, consumer i choosing plan n_i and type ω_i receives the following utility when charging at station $j \in C_i$:

$$
\begin{aligned}
u_{ijt}(p_{j,n_i}, \theta, \omega_i) = & \underbrace{\beta X_j}_{\delta_j} + \underbrace{\beta_{detour}\tau_{ij} - \alpha p_{j,n_i} + \beta_{speed}\kappa_{ij}}_{\mu_{ij}} \\
& + \xi_j + \underbrace{(1-\sigma_{type})\zeta^{fast}_{i,type} + (1-\sigma_{fast})\varepsilon_{ij}}_{\text{Nested logit shocks}}
\end{aligned}
$$

In this equation, δ_j represents the consumer-invariant component of station utility. X_j is a vector of the (non-speed) characteristics of station j. It contains the number of chargers in each kW speed bin ($\{(0, 22], (22, 50], (50, 150], (150, 250], (250, 500]\}$) as well as indicator variables reflecting the major road types near the charger (Autobahn, highway, trunk primary) as well as the presence of nearby amenities (Autobahn service area, fast food, restaurant, convenience store, grocery store, etc.). I also include a network fixed effect.

μ_{ij} represents the individual-specific component of station utility. There are three components: detour time, price, and speed. The detour time τ_{ij} represents the shortest detour required to access station j given i's routes R_i. I first define $\tau(j, r)$ as the shortest detour time (in minutes) required to access station j on route r.[11] An example of these detour times for an example route can be found in Figure 4 for a sampled origin and destination pair. Each dot represents a charging station in the choice set, and the color of the dot corresponds to the detour time. Then, define $\tau_{ij} = \min_{r \in R_i} \tau(j, r)$ as the minimum detour required to access station j over the set of routes R_i. As a given charger may be close to multiple routes of the same consumer, I assume the consumer uses the charger on the trip where the detour is shortest.[12]

The second component of μ_{ij} is the price p_{j,n_i}. p_{j,n_i} refers to the price of charging at station j under their choice of plan n_i. If station j is in-network for plan n_i, then p_{j,n_i} will be the discounted in-network rate. Otherwise, the consumer will pay the basic price to

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[11] For route r characterized by origin and destination pair (O, D), $\tau(j, r)$ takes the form:

$$\tau(j, r) = d(O, j) + d(j, D) - d(O, D)$$

, where $d(A, B)$ indicates the duration of the optimal route between arbitrary points A and B.

[12] This eliminates the need to consider the choice of which route to charge on. Estimating such a model would require information on consumer charger choice conditional on route, which I do not have.

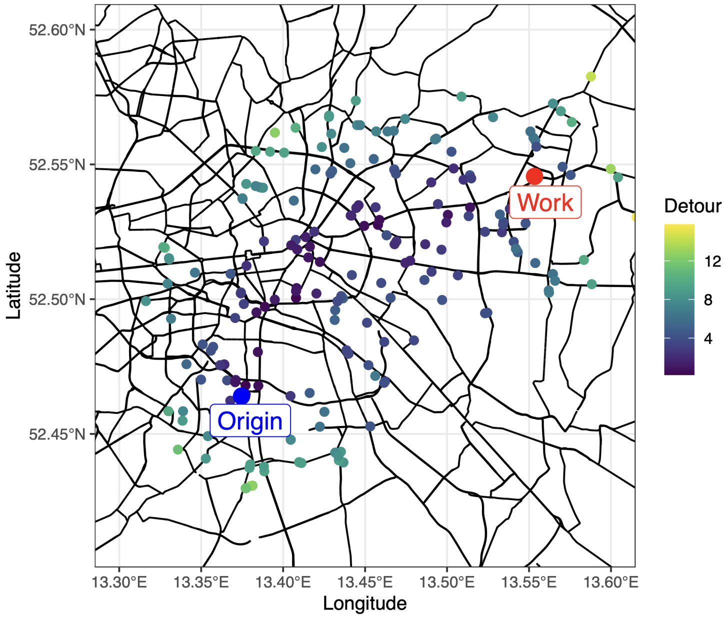

Figure 4: Example of the choice set associated with a consumer's commute. The color of each dot indicates the detour time (relative to the original route) in minutes. Only major roads (i.e. Autobahn and primary highways) are plotted for clarity.

whichever network operates station j. Parameter α represents consumers' sensitivity to per-kWh charging prices. I assume this is common across individuals and do not model it as a random coefficient.

Finally, κ_{ij} represents the effective charging speed realized by consumer i when charging at station j. I define $\kappa_{ij} = \min\{\kappa_i, \kappa_j\}$, where κ_i is the maximum charging rate of consumer i's vehicle and κ_j is the maximum charging speed of the station (both measured in kW). The realized maximum charging speed will be the minimum of these two. As a consequence, consumers with faster charging vehicles will have stronger preferences over charging speed than those with slower charging vehicles.

ξ_j represents the unobserved quality of station j. This is unobserved to the econometrician and could be attributable to a variety of factors which affect consumers' willingness to choose a particular station. Such factors could include unreliable chargers, the station being difficult to access, or the presence of nearby positive amenities not captured in my data.

I assume that ε_{ij}, $\zeta_{i,type}^{fast}$ are IID Type I Extreme Value random variables. σ_{fast} represents the correlation in the utility shocks within the public fast charger nest. σ_{fast} governs the extent to which consumers substitute between fast charging stations. σ_{type} represents the correlation in utility shocks at the charging type choice nest, where the consumer chooses between the inside option (fast charging) and the outside option nest (slow charging). It governs the extent to which consumers are willing to substitute between fast charging and slow charging.

Alternative charging option utilities — If consumers decide not to charge at a fast charging station, they must pick among their available alternative slow charging options. These options include home, workplace, and public slow charging stations. Consumers may only choose from the options which are available to them.

For each charging option $k \in \{h, w, s\}$, the utility of charging option k (if it is available to the consumer) takes the following form:

$$u_{ik}(\theta, \omega_i) = v_k - \alpha p_{ik} + (1 - \sigma_{type})\zeta_{i,type}^{slow} + (1 - \sigma_{slow})\varepsilon_{ik}$$

Each option has a different intercept which represents consumers' mean preference for that charging type. For instance, this allows drivers to prefer home charging to work charging; or, they could prefer either option to public slow charging. Consumers also incur disutility from charging prices. I allow home and work charging prices to be heterogeneous. I simulate home and work charging prices using information from the USCALE survey as well as a 2022 survey of German EV drivers with home charging. I outline the procedure for simulating these prices in the appendix. I normalize the price of public slow charging to be €0.5/kWh, corresponding to the average slow charging price during my sample period. Finally, $\zeta_{i,type}$ and ε_{ik} are IID Type I Extreme Value random variables. σ_{slow} represents the correlation in utility shocks within the outside option nest. It governs the extent to which consumers view the various outside options as close substitutes. As before, σ_{type} represents the correlation in utility shocks at the charging type choice nest.

The availability of home and work charging is determined by the city type the consumer lives in. I sample this according to the empirical probabilities in the USCALE data by city type. Further details on the sampling procedure can be found in the appendix. I assume that all consumers have access to public slow charging.

Station and charging type choice — Assuming that the consumer with plan n_i chooses to charge at a fast charger in their choice set C_i, the probability of charging at station $j \in C_i$ is given by

$$s_{j|F}(p_{n_i}, \theta, \omega_i) = \frac{\exp((\delta_j + \mu_{ij}(p_{j,n_i}, \omega_i) + \xi_j)/(1 - \sigma_{fast}))}{\sum_{k \in C_i} \exp((\delta_k + \mu_{ik}(p_{k,n_i}, \omega_i) + \xi_k)/(1 - \sigma_{fast}))}$$

Similarly, conditional on choosing to slow charge, the probability of slow charging option k is given by

$$s_{k|S}(\theta, \omega_i) = \frac{\exp((v_k - \alpha p_{ik})/(1 - \sigma_{slow}))}{\sum_{\ell \in \lambda_i} \exp((v_\ell - \alpha p_{i\ell})/(1 - \sigma_{slow}))}$$

The choice of whether to charge at a fast charger or a slow charger is made based on the inclusive value of either option. The probability of fast charging is given by

$$s_F(p_{n_i}, \theta, \omega_i) = \frac{\exp(V_F(p_{n_i}, \theta, \omega_i)/(1 - \sigma_{type}))}{\exp(V_S(\theta, \omega_i)/(1 - \sigma_{type})) + \exp(V_F(p_{n_i}, \theta, \omega_i)/(1 - \sigma_{type}))}$$

$$s_S(p_n, \theta, \omega_i) = \frac{\exp(V_S(\theta, \omega_i)/(1-\sigma_{type}))}{\exp(V_S(\theta, \omega_i)/(1-\sigma_{type})) + \exp(V_F(p(n), \theta, \omega_i)/(1-\sigma_{type}))}$$

where the inclusive values of either type of charging take the following form:

$$V_F(p_{n_i}, \theta, \omega_i) = (1-\sigma_{fast}) \log \left(\sum_{j \in C_i} \exp \left(\frac{\delta_j + \mu_{ij}(p_{j,n_i}, \omega_i) + \xi_j}{1-\sigma_{fast}} \right) \right)$$

$$V_S(\theta, \omega_i) = (1-\sigma_{slow}) \log \left(\sum_{k \in \lambda_i} \exp \left(\frac{v_k - \alpha p_{ik}}{1-\sigma_{slow}} \right) \right)$$

The inclusive value of a particular charging event is consequently:

$$V_C(p_{n_i}, \theta, \omega_i) = (1-\sigma_{type}) \log \left(\exp \left(\frac{V_F(p_{n_i}, \theta, \omega_i)}{1-\sigma_{type}} \right) + \exp \left(\frac{V_S(\theta, \omega_i)}{1-\sigma_{type}} \right) \right)$$

Plan choice — Consumers choose whether to subscribe to any of the available plans based on the expected utility of each plan. Given Q_i total charging events, the consumers' expected utility from selecting a given plan n_i can be expressed as:

$$V_n(p_{n_i}, \theta, \omega_i) = Q_i V_C(p_{n_i}, \theta, \omega_i) - \alpha^i_{plan} \varphi_{n_i} + (1-\sigma_{paid}) \zeta^{paid}_{i,n_i} + \zeta^{plan}_{in}$$

Intuitively, the plan's value is the inclusive value for each charging event times the number of times the consumer expects to charge, minus the disutility from the plan fee φ_n. α^i_{plan} reflects the consumers' sensitivity to plan fees. I parameterize α^i_{plan} to be a function of mileage, where $\alpha^i_{plan} = \alpha^0_{plan} + \alpha^{mileage}_{plan}\text{mileage}_i$. This allows consumers with higher mileage to differ in their sensitivity to plan fees relative to low mileage consumers. Finally, if $n = 0$ (corresponding to the consumer not selecting any subscription), p_{n_i} is just the vector of basic prices for each CPO under plan n_i and $\varphi_0 = 0$.

I assume that consumers first pick whether to subscribe to a paid plan, or whether to choose the outside option of no subscription ($n = 0$). Conditional on picking a paid plan, the probability of picking a particular paid plan n_i is

$$s_{n_i|Paid}(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, \omega_i) = \frac{\exp(V_n(p_{n_i}, \theta, \omega_i)/(1-\sigma_{paid}))}{\sum_{n'>0}^{N_{plan}} \exp(V_{n'}(p_{n'}, \theta, \omega_i)/(1-\sigma_{paid}))}$$

The inclusive value of a paid plan is

$$V_{Paid}(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, \omega_i) = (1-\sigma_{paid}) \log(\sum_{n'>0} \exp(V_{n'}(p_{n'}, \theta, \omega_i)/(1-\sigma_{paid})))$$

The probability a consumer picks a paid plan is given by

$$s_{Paid}(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, \omega_i) = \frac{\exp(V_{Paid}(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, \omega_i))}{\exp(V_{Paid}(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, \omega_i)) + \exp(V_0(p(0), \theta, \omega_i))}$$

where V_0 denotes the inclusive value of the outside option (no subscription).

The probability of picking a particular paid plan n is thus

$$s_n(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, \omega_i) = s_{Paid}(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, \omega_i) \times s_{n|Paid}(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, \omega_i)$$

The probability of choosing not to subscribe to any plan is given by

$$s_0(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, \omega_i) = \frac{\exp(V_0(p(0), \theta, \omega_i))}{\exp(V_{Paid}(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, \omega_i)) + \exp(V_0(p(0), \theta, \omega_i))}$$

Aggregate demand — I now discuss how I use each consumer's choice probability to compute their demand for fast charging. In estimation, I assume a time period of a month. I aggregate the number of charging events observed at each station throughout the month. I use this to compute station-level shares of charging events in each market. My consumer choice model expresses choice probabilities at the charging event level. I will outline how I aggregate these event-level choice probabilities into monthly demand for each consumer.

I begin with a consumer i. Suppose that i has chosen plan n. The consumer needs to complete Q_i charging events over the course of the month, which is determined by the routes they take (R_i) as well as the range of their battery. The probability that i charges at station j in their choice set C_i, conditional on having plan n, is the following:

$$s_{ij|n}(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, \omega) = \underbrace{s_F(p_n, \theta, \omega_i)}_{\text{Fast charging}} \times \underbrace{s_{j|F}(p_n, \theta, \omega_i)}_{\text{Choosing station } j}$$

In other words, the probability of charging at station j on a particular charging event conditional on having plan n is the probability of charging at a fast charger (conditional on plan n) multiplied by the probability of picking station j (conditional on fast charging and having selected plan n).

The expected number of charging events that consumer i on plan n does at station j over the course of the month is simply the number of charging events Q_i multiplied by the probability of picking station j given the plan. That is,

$$q_{ij|n}(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, \omega) = Q_i \times s_{ij|n}(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, \omega)$$

This assumes consumers' utility shocks are IID across the different charging events that they must complete.

Finally, the monthly demand for consumer i from plan n is the plan-specific expected demand for j ($q_{ij|n}$) multiplied by the plan choice probability (s_n):

$$q_{ijn}(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, \omega) = s_n(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, \omega) \times q_{ij|n}(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, \omega)$$

I use this to construct the aggregate demand for stations of network m from plan n:

$$D_m(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, n) = \sum_{j \in J(m)} \sum_{i \in I} q_{ijn}(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, \omega_i)$$

where $J(m)$ denotes the set of stations in network m.

I also define market shares (i.e., the share of charging events) for use in demand estimation:

$$s_j(\boldsymbol{p}, \boldsymbol{\varphi}, \theta) = \frac{\sum_{i \in I} \sum_{n=0}^{N_{plan}} q_{ijn}(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, \omega_i)}{\sum_{i \in I} Q_i}$$

Finally, I define $\Lambda_n(\boldsymbol{p}, \boldsymbol{\varphi}, \theta)$ as the mass of drivers on plan n:

$$\Lambda_n(\boldsymbol{p}, \boldsymbol{\varphi}, \theta) = \sum_{i \in I} s_n(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, \omega_i)$$

3.2 Supply side

I now present the supply side of my model. I assume that networks choose prices and plan details (if applicable) to maximize profits from charging in the given period. I assume networks set network-wide prices. All networks must pay a 19% Value Added Tax (VAT), which applies both to the charging price as well as any subscription revenues. Furthermore, I assume Bertrand-Nash competition. I do not model entry or exit of stations. Additionally, while networks have fixed costs and various costs from maintenance, I do not model these since they should not affect static profit maximization in a given period.

Regular networks — Consider first a CPO m which does not offer any paid plans. I assume that marginal costs (before VAT) are identical throughout the network and are equal to c_m. I discuss this assumption later in this section. Network m's profits can be expressed as the following:

$$\Pi_m(\boldsymbol{p}, \boldsymbol{\varphi}) = (p_m/1.19 - c_m) \sum_{n=0}^{N_{plans}} D_m(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, n)$$

Network m has only one choice variable: the basic price available to all consumers (p_m). By changing p_m, networks are controlling the price that all consumers pay to access their network (regardless of plan). Nevertheless, consumers subscribing to paid plans from other providers will have heterogeneous responses since the vectors of other prices they face are specific to their plan.

The first-order condition for p_m is given by the following:

$$\begin{aligned} 0 &= (p_m/1.19 - c_m) \sum_{n=0}^{N_{plans}} \frac{\partial D_m(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, n)}{\partial p_m} + 1/1.19 \sum_{n=0}^{N_{plans}} D_m(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, n) \\ \iff p_m &= 1.19 \cdot c_m - \frac{\sum_{n=0}^{N_{plans}} D_m(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, n)}{\sum_{n=0}^{N_{plans}} \frac{\partial D_m(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, n)}{\partial p_m}} \end{aligned}$$

This yields the standard Bertrand-Nash markup formula.

Plan-offering networks — The maximization problem is more complex for CPOs that offer plans. Instead of just setting one basic price, these CPOs set 1) a basic price, 2) charging prices for each plan, and 3) monthly subscription fees for each plan. I present the equations which characterize joint optimality of these three decision variables. For simplicity, consider a CPO A which offers one paid plan. Let n_A denote the index of A's paid plan. Consumers on a plan $n \neq n_A$ pay the basic price p_A^{Basic} per kWh. Consumers on plan n_A will pay p_A^{Paid} per kWh and monthly fee φ_A.

$$\Pi_A(\boldsymbol{p}, \boldsymbol{\varphi}) = \underbrace{(p_A^{\text{Basic}}/1.19 - c_A) \sum_{n \neq n_A} D_A(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, n)}_{\text{Profit contribution from basic charging}} + \underbrace{(p_A^{\text{Paid}}/1.19 - c_A) D_A(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, n_A)}_{\text{Profit contribution from paid plan } n_A} + \frac{1}{1.19} \underbrace{\Lambda_{n_A}(\boldsymbol{p}, \boldsymbol{\varphi}, \theta) \varphi_A}_{\text{Plan fee revenue}}$$

There are three first-order conditions: p_A^{Basic}, p_A^{Paid}, and φ_A.

$$\begin{aligned} &[p_A^{\text{Basic}}] && 0 = (p_A^{\text{Basic}}/1.19 - c_A) \sum_{n \neq n_A} \frac{\partial D_A(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, n)}{\partial p_A^{\text{Basic}}} + \frac{1}{1.19} \sum_{n \neq n_A} D_A(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, n) + (p_A^{\text{Paid}}/1.19 - c_A) \frac{\partial D_A(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, n_A)}{\partial p_A^{\text{Basic}}} + \frac{1}{1.19} \frac{\partial \Lambda_{n_A}(\boldsymbol{p}, \boldsymbol{\varphi}, \theta)}{\partial p_A^{\text{Basic}}} \varphi_A \\ &[p_A^{\text{Paid}}] && 0 = (p_A^{\text{Basic}}/1.19 - c_A) \sum_{n \neq n_A} \frac{\partial D_A(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, n)}{\partial p_A^{\text{Paid}}} + (p_A^{\text{Paid}}/1.19 - c_A) \frac{\partial D_A(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, n_A)}{\partial p_A^{\text{Paid}}} + \frac{1}{1.19} D_A(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, n_A) + \frac{1}{1.19} \frac{\partial \Lambda_{n_A}(\boldsymbol{p}, \boldsymbol{\varphi}, \theta)}{\partial p_A^{\text{Paid}}} \varphi_A \\ &[\varphi_A] && 0 = (p_A^{\text{Basic}}/1.19 - c_A) \sum_{n \neq n_A} \frac{\partial D_A(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, n)}{\partial \varphi_A} + (p_A^{\text{Paid}}/1.19 - c_A) \frac{\partial D_A(\boldsymbol{p}, \boldsymbol{\varphi}, \theta, n_A)}{\partial \varphi_A} + \frac{1}{1.19} \frac{\partial \Lambda_{n_A}(\boldsymbol{p}, \boldsymbol{\varphi}, \theta)}{\partial \varphi_A} \varphi_A + \frac{1}{1.19} \Lambda_{n_A}(\boldsymbol{p}, \varphi, \theta) \end{aligned}$$

The main insight here is the fact that the CPO internalizes the effect of each variable. For instance, consider what happens if the firm increases p_A^{Basic}. This has two effects: the direct effect is that consumers who don't have A's paid plan will reduce their charging demand for network A. However, following the increase in the basic price, some consumers will choose to subscribe to A's paid plan. Similarly, increasing p_A^{Paid} will decrease demand from n_A plan users and cause substitution to other plans. However, since A gets a higher margin from consumers accessing their network from other plans, they can potentially recapture a portion of lost profits. These additional considerations set these CPOs apart from CPOs which only offer basic plans.

Marginal costs — I assume firms' marginal costs are comprised of two components: the price of electricity and a fixed amount per kWh in GHG quota revenue. I furthermore assume that the price of electricity is common across all stations within a given network. This is not an unrealistic assumption given Germany's integrated electricity market. Since it is one large grid, CPOs can source power for their entire network of stations in aggregate as opposed to having separate contracts with each local utility.

I use publicly available statistics on commercial electricity contracts reported by the German government to infer CPOs marginal costs.[13] In Germany, the electricity rate available to commercial consumers is decreasing in the entity's annual electricity usage. Users with the smallest usage pay the highest prices, whereas those with the highest usage pay the lowest prices. The price is generally determined by determining which usage bin a consumer falls into: for example, 0 to 20 MW, 20 to 500 MW, 500 to 2,000 MW, etc. The full price schedule and further details can be found in Table 16 in the appendix.

I assume that non-utility CPOs pay a commercial electricity rate corresponding to their total (inferred) level of network demand. I aggregate each CPO's demand (assuming an average charge event size of 35kWh) and find the corresponding electricity price based on Table 16. Utility-backed CPOs will have different costs in the sense that they are buying power directly from the wholesale market and supplying it to themselves. This eliminates the markup included in commercial electricity contracts. I assume that their cost of electricity is the wholesale price plus all applicable grid fees, taxes, and levies for the electricity. The effective price paid by a networks includes mandatory taxes and fees but excluding VAT, as networks are eligible for remittance of VAT paid when obtaining electricity.

Finally, I also assume that CPOs earn €0.04/kWh through the sale of GHG quota credits. The German government's Greenhouse Gas (GHG) quota requires that fuel distributors either 1) reduce emissions by a set amount per year or 2) purchase credits in order to meet their reduction quota. Credits are given to EV owners and public charging infrastructure based on their emissions reduction. That is, for each kWh sold by a CPO, they receive a certain amount of credits based on the estimated amount of emissions averted. This depends on the overall energy mix in the grid. CPOs can gain a small amount of revenue by selling their credits to fuel distributors that are not capable of meeting their quota. While the value of these credits fluctuates over time due to quota targets and the grid mix, their value averaged around €0.04/kWh sold during the sample period.[14]) I use this value in my analysis and assume that CPOs' marginal costs are €0.04/kWh lower than the cost of electricity they use.

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[13] https://www-genesis.destatis.de/datenbank/online/url/3d80e83e

[14] https://emobilitaet.online/news/experteninterviews/8451-interview-thg-quote-2026-preise

4 Estimation methodology

4.1 Demand estimation

For each station j in a given market, let D_j denote the observed number of charging events at station j over the course of the month. For ease of exposition, I suppress market subscripts, but it should be understood that all objects are computed separately across markets. Each consumer i in the market must complete Q_i charging events per month, implying a total market size of $\bar{Q} = \sum_{i \in I} Q_i$. I then compute station-specific market shares as $s_j = \frac{D_j}{\bar{Q}}$. This yields a vector of observed charger market shares $\boldsymbol{s}$.

Given an initial value of nonlinear parameters θ, I use the approach of Berry, Levinsohn, and Pakes 1995 to solve for the vector of station mean utilities $\boldsymbol{\delta}$ which rationalize the observed charger market shares $\boldsymbol{s}$. I begin by initializing a guess of $\boldsymbol{\delta}_0$. Given this $\boldsymbol{\delta}_0$, I solve consumers' utility maximization problem and obtain predicted market shares $\boldsymbol{s}_0$. I iteratively update $\boldsymbol{\delta}_t$ to find the solution to the BLP fixed point: $\boldsymbol{\delta}_{t+1} \leftarrow \boldsymbol{\delta}_t + \log(\boldsymbol{s}) - \log(\boldsymbol{s}(\boldsymbol{\delta}_t))$. I use the SQUAREM quasi-Newton algorithm to accelerate the standard BLP contraction mapping. The procedure is described below:

$$\boldsymbol{\delta}_{t+1} \leftarrow \boldsymbol{\delta}_t - 2\alpha_t \boldsymbol{r}_t + (\alpha_t)^2 \boldsymbol{v}_t$$

where

$$\alpha_t = \frac{(\boldsymbol{v}_t)' \boldsymbol{r}_t}{(\boldsymbol{v}_t)' \boldsymbol{v}_t}, \quad \boldsymbol{r}_t = f(\boldsymbol{\delta}_t) - \boldsymbol{\delta}_t, \quad \boldsymbol{v}_t = f(f(\boldsymbol{\delta}_t)) - 2f(\boldsymbol{\delta}_t) + \boldsymbol{\delta}_t$$

and $f : \boldsymbol{\delta}_{t+1} \leftarrow \boldsymbol{\delta}_t + \log(\boldsymbol{s}) - \log(\boldsymbol{s}(\boldsymbol{\delta}_t))$ represents the standard BLP contraction mapping. I iterate until $\|\boldsymbol{\delta}_t - \boldsymbol{\delta}_{t-1}\|_\infty < \epsilon$, where $\epsilon =$`1E-8`.

I then project $\boldsymbol{\delta}$ on station and location characteristics as well as a network fixed effect, with m representing the CPO operating station j:

$$\delta_j = \beta X_j + \nu_m + \xi_j$$

I then use the residual $\xi_j(\theta)$ interacted with a number of instruments Z_j to form the station demand moments:

$$g_{\text{dem}}(\theta) = Z_j' \xi_j(\theta)$$

For instruments, I find all chargers within 100 m, 333 m, 1 km, and 5 km of each charging station. Within each ring, I compute the number of stations, the number which offer rapid charging (speed greater than 150 kW), the average station size, and the average station maximum speed. These instruments will be proxies for the presence of local competition, as well as the average quality of these competitors.

4.2 Micro moments

I construct a number of moments using the consumer micro data.

I construct moments based on public fast and slow charging shares for each type. Additionally, I include moments for the work charging share for type hw. Since fast and slow charging are already included for type w, including the work charging share for them would be redundant. Similarly, including home charging shares would be redundant for types h and hw. I also construct moments based on the standard deviation of the public charging share for each type. This disciplines the model to better match the shape of the public charging share distribution in the data. For all type-specific moments, I weigh by the proportion of consumers who have each type.

In the micro data, I observe the work charging share conditional on the binned price difference between work and home charging. I construct a separate moment for each price difference bin. I construct a moment matching the observed work charging share for all consumers with a (simulated) work price difference within each bin. I also construct a moment based on the proportion of consumers who select a paid plan from each provider. As I unfortunately don't see this conditional on type, these moments are not conditional on type.

I form the micro moment vector $g_{micro}(\theta)$ by stacking all of these moments together.

4.3 Supply side

On the supply side, for CPOs which do not offer plans, I invert their first-order condition to obtain the model-implied marginal costs. I only do this for the networks which do not offer subscription plans. I then compute the difference between the implied marginal cost and the commercial (or wholesale, for utilities) electricity price corresponding to the network's level of aggregate demand, net of the GHG quota credit value. I stack these into the moment vector $g_{mc}(\theta)$:

$$g_{mc}(\theta) = \begin{bmatrix} \hat{c}_1(\theta) - c_1 \\ \vdots \\ \hat{c}_{N_m}(\theta) - c_{N_m} \end{bmatrix}$$

To be clear, $g_{mc}(\theta)$ does not contain entries for CPOs that offer plans. Those networks are dealt with separately.

For the networks which offer plans, I construct moments based on the first-order conditions for each of their decision variables (basic price, paid plan prices, and plan fees), assuming that their marginal cost is equal to the electricity prices corresponding to their aggregate electricity usage (and utility vs. non-utility status), net of the GHG quota credit value. This gives me three sets of moments for each plan-providing CPO: $f^{\text{Basic}}(\theta)$ (FOC for the basic fee), $f^{\text{Paid}}(\theta)$ (FOC(s) for the plan price(s)), and $f^{\varphi}(\theta)$ (FOC(s) for plan fee(s)). I stack these up for all providers to form $g_{FOC}(\theta)$. As an example with CPOs A and B and one plan tier for each:

$$g_{FOC}(\theta) = \begin{bmatrix} f_A^{\text{Basic}}(\theta) \\ f_A^{\text{Paid}}(\theta) \\ f_A^{\varphi}(\theta) \\ f_B^{\text{Basic}}(\theta) \\ f_B^{\text{Paid}}(\theta) \\ f_B^{\varphi}(\theta) \end{bmatrix}$$

4.4 GMM objective

The moment vector used for estimation is formed by stacking each of the separate moment vectors (demand moments, micro moments, marginal cost moments, and the FOC moments):

$$g(\theta) = \begin{bmatrix} g_{dem}(\theta) \\ g_{micro}(\theta) \\ g_{mc}(\theta) \\ g_{FOC}(\theta) \end{bmatrix}$$

The GMM objective function is formed using optimal weighting matrix $W = S^{-1}$, where S indicates the variance-covariance matrix of $g(\theta)$:

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}}\, g(\theta)' W g(\theta)$$

I search for the θ which minimizes the objective using Nelder-Mead.

4.5 Identification

There are two parameters which are central to the analysis: the disutility of detours and price sensitivity. The detour coefficient is identified using the logic behind Berry, Levinsohn, and Pakes 1995 and Houde 2012. The differentiation instruments are constructed using the number and quality of competing stations within rings ranging from 100 meters to 5 kilometers around each station. The rate at which demand falls as the distance between stations increases is governed by how much consumers dislike detour times. If consumers are unwilling to take large detours, then the presence of local competitors will have a much larger impact on a station's demand than distant competitors. Conversely, if consumers are willing to tolerate long detours, then the presence of rival stations at longer distances will potentially affect demand for the station. There is sufficient cross-sectional variation in the presence of local competition to identify this parameter.

The identification of the price coefficient is more difficult. This is because prices do not vary over time and are uniform across each network. This rules out standard techniques for handling the endogeneity of prices. Instead, I identify the price coefficient primarily using firms' first-order conditions for prices and plan fees. I assume that each firm faces a marginal cost equal to the commercial price of electricity corresponding to its volume of electricity demand, including any mandatory taxes and fees and net of the value of greenhouse gas quota credits. For networks operated by utility companies, I assume they face a marginal cost equal to the wholesale price of electricity (including taxes and fees), net of the greenhouse gas quota credits.

For the networks which only set a single linear price, the supply-side moments require that the marginal cost implied by the first-order conditions match the firm's effective electricity price. For subscription networks which set both prices and fees, the moments require that the first-order conditions for the basic price, the plan price, and the plan fee all hold at the firm's electricity price. The price coefficient must rationalize both the linear prices set by non-plan networks as well as the discounted plan prices.

I also use additional micro moments from the USCALE data to help identify the price coefficient. Specifically, I target the share of charging done at work as a function of the price difference between home and work charging. The survey contains both the price difference between home and work charging (binned in ten cent intervals) as well as the share of charging done at either option. The share of work charging is higher when work charging is cheaper and decreases as work charging becomes costlier. Requiring that my model match how sensitive consumers are to the relative price difference between these options helps identify their price sensitivity without relying solely upon the supply-side.

5 Results

I present the estimated nonlinear parameters in Table 5. All of the coefficients have the intuitive sign: drivers have downward sloping demand, prefer faster chargers, and rank home, work, and public slow charging in the same order observed in the survey data.

Coef	Estimate	SE
$\hat{\alpha}$	0.478	0.009
$\hat{\alpha}_{plan}$	0.227	0.009
$\hat{\alpha}_{mileage}$	-0.547	0.026
$\hat{\beta}_{detour}$	-0.332	0.023
$\hat{\beta}_{speed}$	0.136	0.027
$\hat{\sigma}_{fast}$	0.856	0.002
$\hat{\sigma}_{slow}$	0.813	0.007
$\hat{\sigma}_{type}$	0.840	0.005
$\hat{\sigma}_{paid}$	0.458	0.037
$\hat{v}_h$	-1.699	0.153
$\hat{v}_w$	-1.842	0.152
$\hat{v}_s$	-2.039	0.152

Table 5: Nonlinear parameter estimates

5.1 Elasticities

Station-level price elasticities — Without context, it is difficult to jointly interpret the parameter estimates in Table 5. One of the key questions is whether the model delivers reasonable price elasticities. To address this, I present station-level price elasticities in Table 6. These are computed by assuming that charging prices at all levels (both basic and plan) change (rather than, for instance, only changing the basic price while keeping the paid plan price fixed). I find an average price elasticity of -1.79, suggesting that consumers are quite responsive to price. The standard deviation is 0.37, indicating that there is substantial heterogeneity at the station level. These results are in line with those from Bernard et al. 2025, who document short-run elasticities ranging from -2.9 to -2 in a large-scale randomized control trial for UK public fast chargers.

Outside option	Mean	SD	p10	p25	p50	p75	p90
Home & Work	-1.88	0.36	-2.34	-2.05	-1.92	-1.69	-1.39
Home	-1.84	0.37	-2.30	-2.04	-1.90	-1.62	-1.32
Work	-1.85	0.43	-2.34	-2.06	-1.93	-1.66	-1.35
Neither	-1.73	0.42	-2.23	-1.99	-1.81	-1.48	-1.20
Aggregate	-1.79	0.37	-2.25	-2.01	-1.85	-1.57	-1.28

Table 6: Station-level elasticities, presented in aggregate as well as by consumer type (hw, h, w, n).

I also break down elasticities by consumers' availability of outside option. Perhaps not surprisingly, hw consumers are the most elastic (-1.88), and n consumers are the least elastic (-1.73). Consumers with the largest set of alternative options are the most sensitive to public charging price since their other options are cheaper and more convenient. Type n consumers, who have no alternative other than public slow charging, are the least sensitive with regard to price.

Cross-price elasticities — Consumers are more likely to substitute between nearer chargers than between farther ones. In Figure 5, I plot the average diversion ratio as a function of distance (km) between stations. I find that cross price elasticities decay quickly in the distance to the rival station, as expected. This is a composition of two effects: 1) nearby stations are likely to have more consumer choice sets in common, and 2) given that two chargers are in the same choice set, consumers will be less likely to substitute to station that requires a greater detour from their route (all else held equal).

I also document heterogeneous preferences for charging speed based on the maximum charging speed of consumers' vehicles. In Figure 6, I plot station price elasticities as a function of the station's maximum power rate (in kW), separately for each vehicle charging power bin. Consumers with slower charging vehicles have less elastic demand for slower chargers and more elastic demand for faster chargers. The relationship flips for faster charging con-

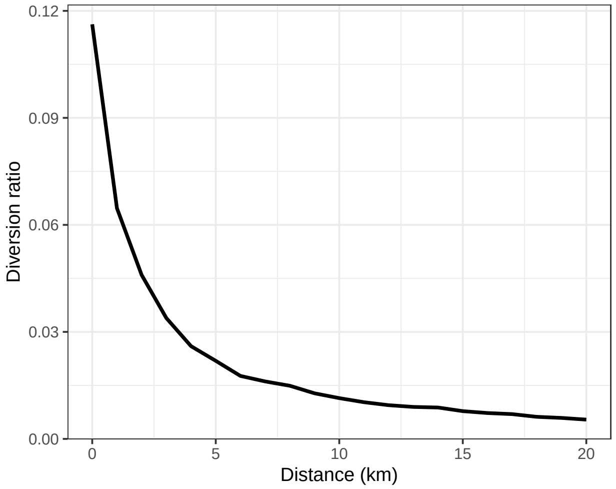

Figure 5: Mean diversion ratio as a function of distance between stations. The mean diversion ratio within each 500 meter distance bin is plotted.

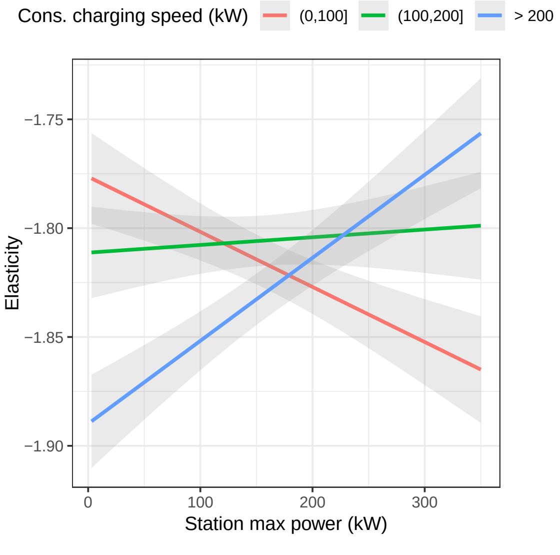

Figure 6: Station price elasticity by station max charging power (kW) and consumer binned maximum charging speed (0 to 100 kW, 100 to 200kW, and greater than 200 kW). For each consumer charging speed bin, I regress the station price elasticity on the station's max power.

sumers: they have more elastic demand for slower chargers and less elastic demand for fast chargers. This relationship is expected. Consumers with fast charging vehicles experience time savings from faster chargers and are willing to pay more for them, while consumers with slow charging vehicles gain virtually no benefit from faster charging.

Variation in vehicle charging speed creates heterogeneous preferences for speed and, therefore, heterogeneous substitution patterns. Vehicle charging speed controls the level of perceived quality differentiation between charging stations and networks. Faster charging consumers have stronger preferences for speed and therefore tend to pick faster stations. Slower charging consumers experience little variation in charging speed, causing them to care less about speed differences in their station choice.

5.2 Plan choice and usage

Which types of consumers are more likely to choose a plan? The answer is 1) consumers without access to charging at home or work and 2) consumers who drive a lot. Table 7 reports the aggregate paid plan share within each mileage bin (in 10 thousand km bins) and outside option availability. Consumers without charging at home or work are more likely to subscribe to a plan across all mileage types. In general, the aggregate paid plan share is higher for types which rely more on public charging. It is also increasing in mileage. As the required number of charging events increases, so too does the incentive to subscribe to a plan.

Mileage	hw	h	w	n
$(0, 10000]$	0.07	0.07	0.09	0.11
$(10000, 20000]$	0.20	0.20	0.24	0.31
$(20000, \infty]$	0.56	0.57	0.63	0.75

Table 7: Overall plan takeup based on consumer outside option availability and annual mileage bin, aggregating choice probabilities across all paid plans. Mileage is grouped into bins of 10,000 km.

I also compare the charging behavior of plan subscribers with non-subscribers. By design, subscription plans target heavy charging consumers. A subscription only makes sense if a consumer expects to do enough public charging within the selected network to break even. My model predicts that the consumers who select plans will, on average, be those who rely more on public charging.

I find that this is indeed the case: subscribers use public charging far more than non-subscribers. The average consumer does 0.86 charging events per month, whereas the average plan consumer does 2.44 charging events per month. Breaking this down by plan, I find that the average EnBW subscriber does 2.74 events per month, Ionity subscribers do 3.34 per month, and Aral subscribers do 1.79 per month. The magnitude of the difference is related to the magnitude of the discount that each plan gives: Aral has the smallest discount

(€ 0.04/kWh), followed by EnBW (€ 0.10/kWh) and Ionity (€ 0.20/kWh). I also find that plan consumers do a greater share of charging within their selected network than they would otherwise. Compared with their in-network shares when not subscribed, in-network shares are 24.6% higher for EnBW, 66.8% higher for Ionity, and 12.1% higher for Aral.

Network	No plan	Own plan
EnBW	-1.74	-1.33
Ionity	-2.05	-1.18
Aral Pulse	-1.82	-1.62

Table 8: Average station-level price elasticities by plan network for consumers 1) who are not subscribed to any plan and 2) on the network's own plan.

Consumers on subscription plans are also more loyal to their chosen network. Table 8 summarizes the station-level price elasticities by network for both non-subscribers and plan subscribers. The no plan column reports the average station-level price elasticity across consumers who do not subscribe to any paid plan. The own plan column does the same but for consumers who are on a network's plan. I find that plan consumers are significantly less elastic compared with non-plan consumers. The difference is largest for Ionity (-1.18 vs. -2.05) and smallest for Aral (-1.62 vs -1.82). The primary reason for the difference is likely the magnitude of difference between each plan price and the corresponding non-subscriber price. This provides evidence that plans alter consumers' charging behavior and incentivize them to do more charging within the network of their plan.

Plan consumers also substitute more strongly to in-network stations than do non-subscribers. To document this, I compute station-level cross-price elasticities separately for each plan. That is, for a given station, I model the effect of a price change at that station under each plan separately. I compute the change in plan-specific demand for that station (and its competitors) in response to the price change. I compute this conditional on plan choice to isolate substitution within public charging from substitution across plans. Using these price and cross-price elasticities, I compute diversion ratios which I aggregate to the network level. Table 9 presents the within-network diversion ratios for each plan network, broken down by plan membership. In response to a price increase at an in-network station, plan subscribers

Network	No plan	Own plan
EnBW	0.082	0.172
Ionity	0.003	0.038
Aral Pulse	0.039	0.061

Table 9: Within-network diversion ratio by plan network for consumers 1) who are not subscribed to any plan and 2) on the network's own plan. Diversion ratios to each of the network's other stations are aggregated.

Network	No plan	Own plan
EnBW	0.072	0.131
Ionity	0.006	0.051
Aral Pulse	0.042	0.072

Table 10: Average (aggregate) diversion ratio from other networks to plan networks for consumers 1) who are not subscribed to any plan and 2) on the network's own plan. Diversion ratios to each of the plan networks' stations are aggregated.

are far more likely to substitute to a different in-network station than non-subscribers. For example, EnBW subscribers are more than twice as likely to substitute to another EnBW station in response to a price increase than non-subscribers are. Similar patterns hold for Ionity and Aral Pulse. It should be noted that Ionity's diversion ratios are lower. This is mechanical: they have fewer stations and their network is spread out along the highway network. Diversion ratios from other providers' stations to plan networks' stations are also higher for plan consumers. Table 10 presents the average diversion ratio from other networks to each of the three plan-offering networks, broken down by plan membership. When facing a price increase at an out-of-network station, plan subscribers are far more likely to substitute to an in-network station than non-subscribers facing the same price increase.

In summary, subscription plans cause consumers to be less price sensitive in-network, more likely to substitute within network, and more likely to substitute from competitors' stations to their own. These three facts change the nature of competition for this consumer segment. Plans increase loyalty and decreasing willingness to substitute out-of-network. Consequently, they make it more difficult for other firms to attract plan consumers.

5.3 Goodness of fit

My model matches the observed variation in consumer charging behavior which I observe in the data. Figure 7 compares model-predicted charging shares for each charging option with their sample analogue by consumer outside option availability. The model is able to match charging shares for each type of charging quite well, both for fast charging as well as each of the outside options. It explains much of the heterogeneity in charging behavior across types.

Table 11 compares model-predicted fast charging shares for different consumer mileage bins with those observed in the survey data. Even though these moments were not targeted in estimation, the model is able to rationalize the increased share of public fast charging as mileage increases and how it varies with type. The model does especially over-predict fast charging shares for low mileage h consumers (12% compared with 8% in the data), but the disparity decreases as mileage increases. It also over-predicts for n type consumers with mileage between 10 and 20 thousand kilometers, but this is mainly because of the non-monotonicity in the micro data itself.

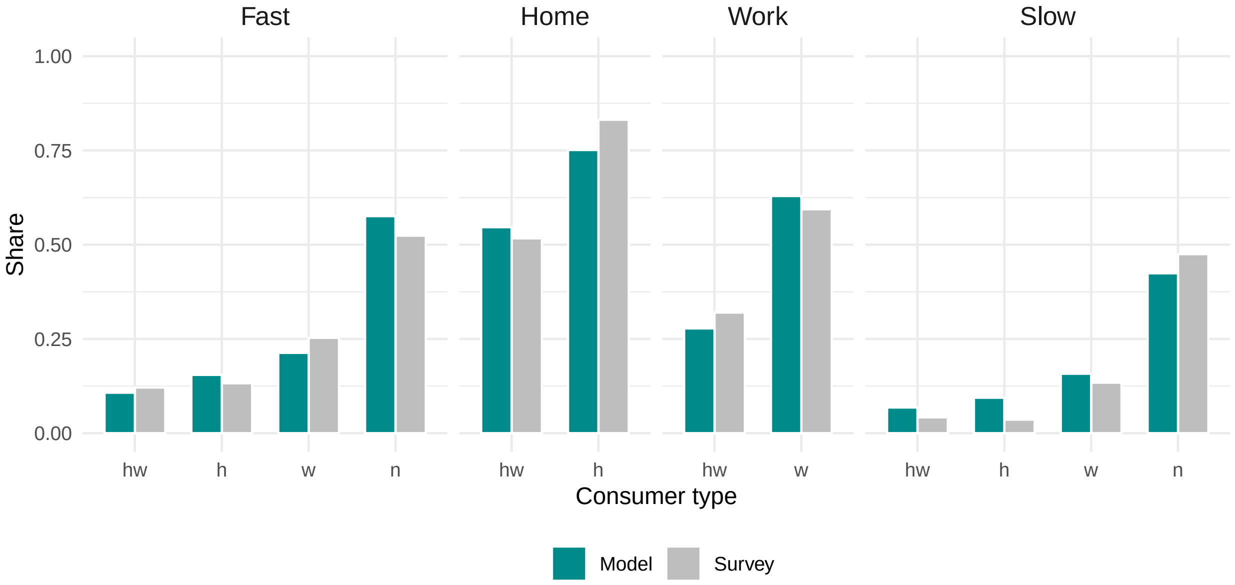

Figure 7: Comparison of model-predicted and empirical charging shares at each charging type (fast, home, work, slow) across each consumer type (hw, h, w, n). The model-predicted shares are indicated in cyan, while their sample analogues are indicated in grey. Each group of bars corresponds to a given consumer type. The type of charging is indicated at the top of each plot panel.

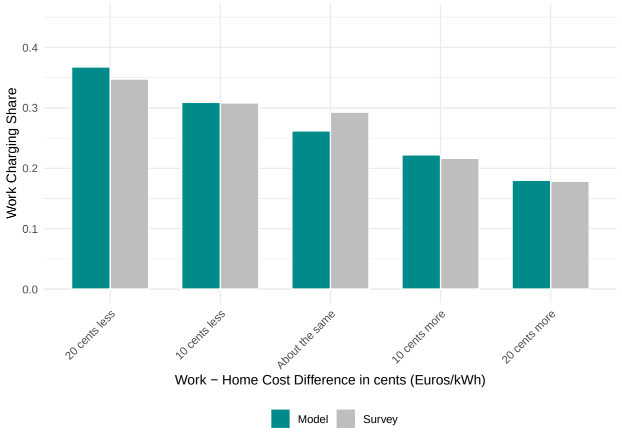

Figure 8: Work charging share as function of the price difference (in cents €/kWh) between work and home charging costs. The cyan bars correspond to the model-predicted work shares, while the grey bars correspond to their empirical analogues in the micro data. Each set of bars corresponds to the difference between work and home charging costs, binned in 10 cent increments. The comparison is done on the subset of consumers who have charging both at home and at work ($N = 395$).

	hw		h		w		n	
Mileage	Survey	Model	Survey	Model	Survey	Model	Survey	Model
$(0, 10000]$	0.08	0.09	0.08	0.12	0.20	0.17	0.54	0.50
$(10000, 20000]$	0.12	0.11	0.14	0.16	0.24	0.22	0.48	0.60
$(20000, \infty]$	0.14	0.15	0.19	0.20	0.32	0.29	0.63	0.66

Table 11: Share of model-predicted public fast charging by consumer type and binned annual mileage, compared with observed public fast charging shares in the survey.

Finally, Figure 8 compares the predicted work shares as a function of the difference between (simulated) work and home charging prices with their sample analogues. In the micro data, I observe that the share of charging done at work decreases as work charging becomes more expensive relative to home charging. The model is able to replicate this relationship well and predicts a similarly declining share of work charging. This demonstrates that the estimated price coefficient and nesting parameters generate plausible substitution patterns for the outside option.

6 Counterfactual analysis

The goal of this paper is to decompose the contributions of network structure and price discrimination to market power. To this end, I simulate the effect of two different policies. First, I ban price discrimination and assume that each CPO may only set one linear price for all consumers. This shuts down the plans offered by the plan-offering networks and puts all networks on even footing. I then compute the equilibrium prices resulting from this new assumption on market structure.

The second policy randomly changes network structure by removing various numbers of stations from plan-offering networks (EnBW, Ionity, and Aral) and reallocating them at random to their competitors. I do this separately for each network. I assume that the plan networks still engage in price discrimination using subscription plans. This scenario examines how networks' profits from subscription pricing vary as their density and route coverage are degraded.

Finally, I simulate the joint effect of these policies. By comparing the simultaneous effect of banning price discrimination *and* removing stations from plan networks, I can quantify the complementarities between subscription pricing and network differentiation.

6.1 Banning price discrimination

In this counterfactual scenario, I ban price discrimination by shutting down subscription plans. Instead, all CPOs must set one linear price for all consumers. I solve for equilibrium prices by iterating on the pricing FOC equation until convergence:

$$\boldsymbol{p} \leftarrow \boldsymbol{c} + \eta(\boldsymbol{p})$$

where $\boldsymbol{c}$ represents the vector of CPO costs and $\eta(\cdot)$ represents the vector of markups. Banning price discrimination causes a reduction in prices industry-wide. The sales-weighted average price decreases by 9%. Plan-offering networks decrease prices by 5.9%, while non-plan networks decrease their price by 10.7%. The aggregate quantity of charging increases by 5.5%, with quantity falling by 4% for plan-offering CPOs and increasing by 13.4% for non-plan CPOs. This represents a reallocation of demand between plan and non-plan networks. Finally, profits fall for both plan-offering and non-plan networks. Profit falls by 18.2% for plan-offering CPOs, driven by both the reduction in the linear price and lost plan fee revenue. Aggregate profit for non-plan CPOs falls by 3.2% with the reduction in charging prices offsetting increased demand.

On the consumer side, I find that the vast majority of consumers benefit from the elimination of price discrimination, while a small number of consumers are worse off under the policy. This is a bit counter-intuitive: we would expect that consumers who subscribed to a network would be worse off after the elimination of price discrimination. Instead of paying the discounted rate, they must now pay the network's higher regular rate. However, two effects explain this result. The first is the fact that once plans are shut down, plan-offering CPOs decrease their linear prices. This lowers the magnitude of the price increase the subscriber faces. The second is the fact that other networks respond to the plan elimination by also reducing their prices. When consumers re-optimize their demand, they are able to lower their charging costs.

Why do subscription plans end up raising prices? The intuition is that subscription plans segment the market into heavy versus light chargers and make it harder for regular networks to contest the heavy charging segment. By design, plans make subscribers more loyal to their chosen network and reduce their willingness to substitute to other networks. If rival networks wanted to attract such consumers, they would have to cut prices significantly. However, doing so would also reduce their margin on existing consumers. By revealed preference, regular networks are better off serving the light charging segment with higher margins than trying to serve both segments with lower margins.

Things change once plans are eliminated. First, plan networks reduce their linear price. Under subscription pricing, they set a linear price higher than they would under uniform pricing to steer consumers towards their plan. Once plans are eliminated, their linear price will fall. Second, the heavy charging segment now becomes more contestable as they no longer face discounted in-network prices. Regular networks thus decrease their prices to attract this segment which was previously made difficult to access because of the subscription discounts. These two effects both put downward pressure on linear prices.

6.2 Network size

I now examine how network size interacts with the ability to price discriminate. To do so, I take a plan-offering network and randomly remove a share of its stations, which I reallocate at random to its competitors. For each network, I compute results across four different removal scenarios: 10, 25, 50, and 75% of stations, leaving 90, 75, 50, and 25% of the network's original stations. Using each resulting ownership structure, I compute equilibrium prices and plan fees. I then shut down subscription plans and solve for equilibrium linear prices. This yields a counterfactual for each network size, both with and without subscription pricing. I repeat the analysis for each network and network size combination 25 times to account for sampling variability.

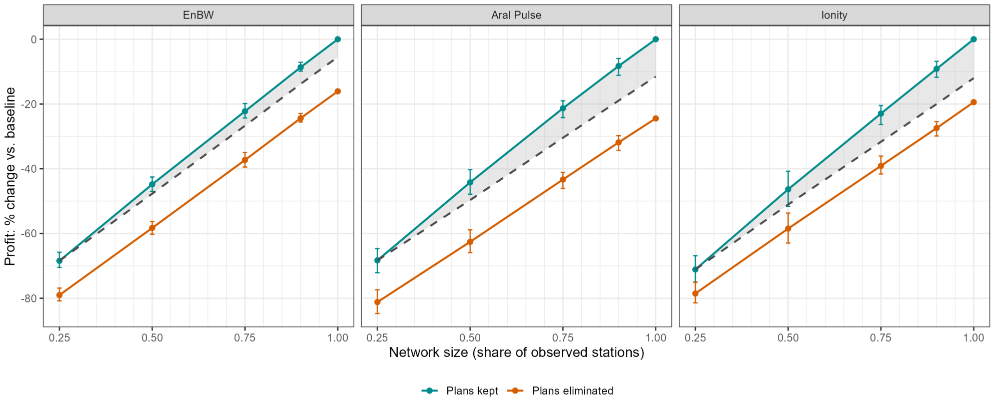

Figure 9: Profits under counterfactual network size and plan elimination status. The horizontal axis represents relative network size: 0.25 corresponds to a network which is 25% of the observed network, while 1 corresponds to the observed network. Profits under subscription pricing are plotted in teal, while profits under plan elimination are plotted in orange. The grey dashed line illustrates a constant gain from network size, while the shaded region represents the complementarities between subscription and network size.

Figure 9 shows the results the counterfactual for each plan-offering network. Network size strongly affects profitability, and the relationship is close to linear. Eliminating subscriptions further lowers profits at every network size. The dashed line shifts the no-plan profit line up by the gain from subscription pricing observed when the network is 25 percent of its actual size. If the gain from subscriptions did not depend on network size, the plan and no-plan lines would be parallel. Instead, the gap between them widens as the network grows, and the shaded region captures the additional gain from subscriptions that comes through network size. This is consistent with complementarities between network size and the ability to price discriminate.

To investigate this further, Figure 10 plots the gain from subscription pricing in absolute terms against actual network size, with all three networks on the same graph. The three networks trace out a common frontier. The gain from subscription is increasing and concave in network size. This suggests that the marginal contribution of additional stations falls once

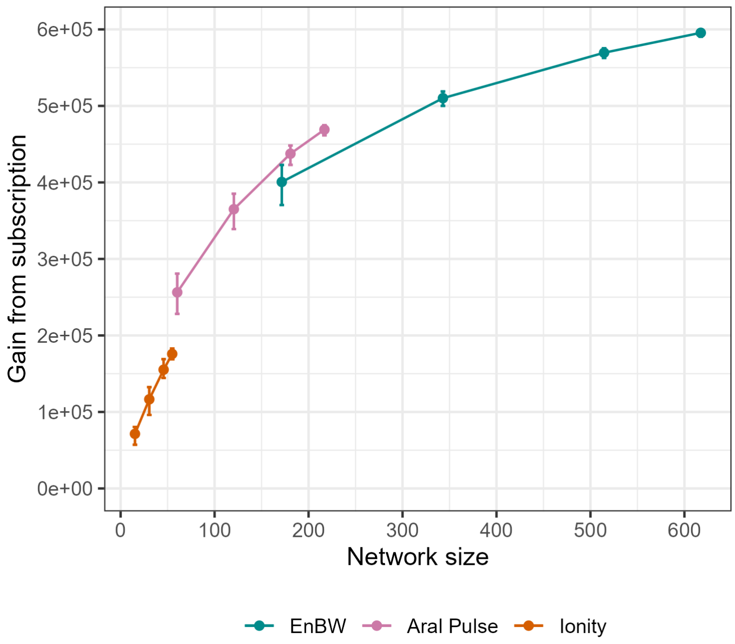

Figure 10: Gain from subscription plans as a function of network size for each of the three subscription networks. The horizontal axis represents the number of stations in the network, while the vertical axis represents the gain in profits from offering subscription pricing.

the network reaches sufficient density.

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A Additional USCALE survey descriptive statistics

A.1 Share of home and work charging by city type

City type	P(Home)	P(Work)
Large > 1M	0.69	0.47
Large > 500k	0.74	0.41
Large >100k	0.75	0.36
Medium	0.80	0.27
Suburbs	0.84	0.27
Small	0.86	0.25
Countryside	0.94	0.19

Table 12: Proportion of USCALE survey respondents with home and work charging, conditional on city type.

A.2 Fast charging shares by mileage

Mileage bin (km)	hw	h	w	n	N
(0, 10,000]	0.08	0.08	0.20	0.54	459
(10,000, 20,000]	0.12	0.14	0.24	0.48	961
(20,000, ∞)	0.14	0.19	0.32	0.63	458
N	421	1142	129	186	1878

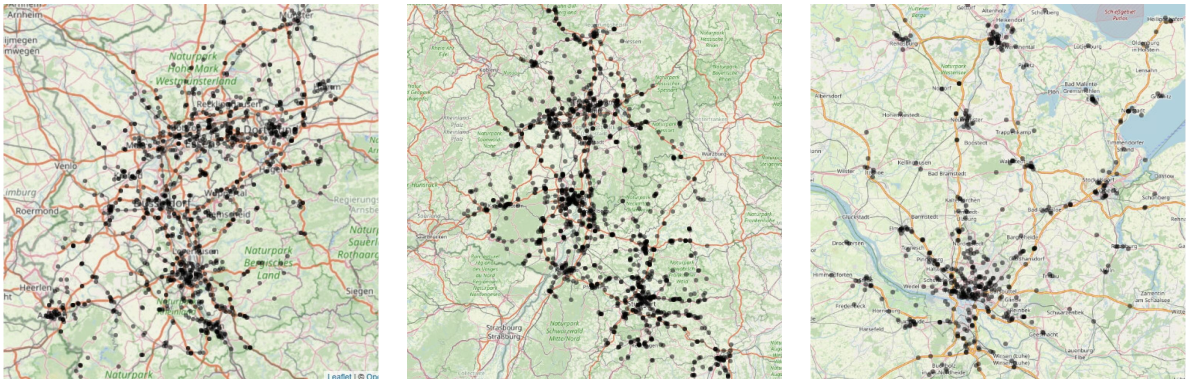
Table 13: Public fast charging share, by consumer outside option type and annual mileage bin. Mileage is binned into 10,000 km bins. Values refer to the share of public fast charging done by consumers of the indicated type and annual mileage bin. The bottom row indicates the number of consumers of each type, whereas the final column indicates the number of consumers in each mileage bin.

A.3 Work charging shares as function of work-home price difference

	Work - home price (cents/kWh, binned)				
	20 cents less	10 cents less	About same	10 cents more	20 cents more
Work share	0.35	0.31	0.29	0.22	0.18
N	56	103	143	64	29

Table 14: Work charging share as a function of the (binned) difference between work and home charging prices (€ cents/kWh). The data comes from type hw consumers who have charging at both home and work. The first row represents the average share of charging done at work, while the second row represents the number of consumers with the designated price difference bin.

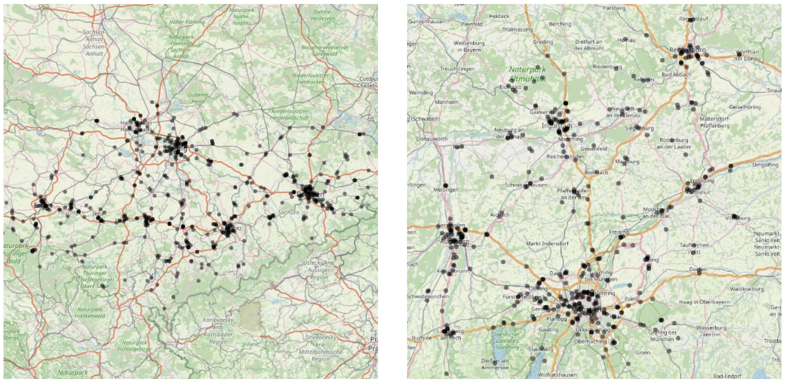
B Stations in sample



(a) Rhine-Ruhr Metropolitan Region

(b) Hessen and Baden-Württemberg

(c) Hamburg Metropolitan Region



(d) Central German Metropolitan Region

(e) Munich Metropolitan Region

Figure 11: The five markets I use in estimation as of September 2024. Each dot corresponds to a charging station.

C Data construction

C.1 Simulating consumer types

Spatial distribution of consumers — I use the KBA vehicle registration data to sample EV drivers. Specifically, using the KBA's FZ 27 dataset, I observe the stock of electric vehicles in each zipcode. I construct 20 representative consumers in each zipcode. If there are N drivers in the zipcode, each simulated driver is given a weight of $\frac{N}{20}$.

Route and choice set construction — I now outline my process for sampling commuting routes. For a given origin region, I use the commuting flow data to determine the probability that a consumer commutes to each destination region. I repeat this for all origin and destination combinations with a commuting distance less than 100 km to obtain the full distribution of commuting paths.

Since the commuting data does not differentiate by mode of travel, I use the Mobility in Germany data to determine the probability of a given commuting trip being by car, conditional on the trip distance and origin region type. I use the sampling weights provided in the survey to ensure the results are representative. I model this using a logit regression with distance as a covariate. I conduct separate regressions for each region type.

$$P(car \mid dist, region = r) = \frac{\exp(\beta_0^r + \beta_1^r \text{dist})}{1 + (\exp(\beta_0^r + \beta_1^r \text{dist}))}$$

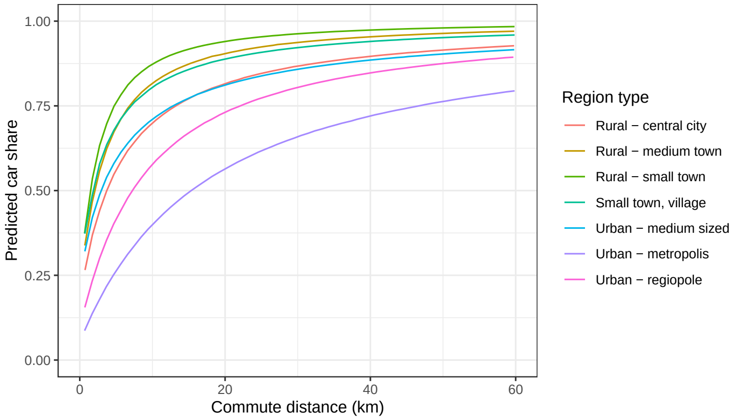


Figure 12: Predicted share of commuting trips taken by car as a function of trip distance and origin region type

The resulting probabilities are plotted in Figure 12. The results are as one may expect: the probability of commuting via car is increasing in trip distance and decreasing in urbanization. Based on these estimates, I then back out the probability each commuting trip was taken via car using Bayes' rule. After taking this into account, I can then sample routes from the (inferred) distribution of car commuting trips.

For each sampled consumer, I sample a commuting path from the distribution of car commuting trips originating in the region. This relies on the assumption that commuting behavior is homogeneous across zipcodes within a municipality. This is a necessary assumption since I do not have finer commuting flow data. Each commuting path is assumed to begin at the centroid of the zipcode. The destination location is sampled in a sequential manner. First, the destination municipality is sampled according to the distribution of vehicle commuting trips. Then, within a destination municipality, I sample the destination according to the empirical distribution of workplaces obtained from the OSM data.

Non-commuting trips are sampled in a similar way. First, I use the Mobility in Germany data to find the distribution of non-commute trip distance conditional on using a car as the mode of transportation. I use the sampling weights provided in the survey to ensure the results are representative. I condition this on the drivers' region type. I discretize this distribution into 5 km bins. In order to simulate non-commuting routes, I first sample a distance bin (e.g., 15 to 20 km). I then sample the destination point based on the distribution of business establishments and points of interest within this distance band (again sourced from the OSM data).

For each sampled trip, I find the route through the road network which minimizes the time to reach the destination. To do this, I utilize the `osmnx` Python package (Boeing 2025), which uses Dijkstra's algorithm and accounts for the driving speed on different road types. As a necessary simplification, I assume traffic is negligible. I then find all charging stations within 5 km of the resulting route. I define this as the consumers' choice set. For each charger within the choice set, I compute the travel time for both 1) the trip from the origin to the charger and 2) the trip from the charger to the destination. These are again found using Dijkstra's algorithm. This allows drivers to re-optimize their path (without having to return to the original path). The detour time is computed as the difference between the total travel time associated with accessing the charger and the original travel time (driving directly to the destination). An example of this can be found in Figure 4.

Each of the sampled consumers in each zipcode is assigned one commuting route and four non-commuting routes. I assume they take the commuting route every weekday and take each non-commuting route once a week. I then compute the mileage implied by these trips, which will be used to determine the number of charging events they need.

For all chargers which appear in at least one choice set, I compute the minimum detour required to access each charger across all of the driver's routes. That is, if a given charger requires a seven minute detour on trip A but only a three minute detour on trip B, I use a detour time of three minutes when computing consumer's utility from charging at that station. This simplification prevents me from having to model both the choice of which route to charge on as well as which charger along that route to select. As I lack the highly granular data that would allow me to model route choice, this simplification is necessary.

Vehicle characteristics — I assign vehicles to simulated consumers based on the observed market shares of each vehicle (in the existing vehicle fleet). However, I need to be more careful than randomly assigning vehicles to consumers independent of their driving behavior. Drivers' choice of EV is endogenous and is likely correlated with their driving behavior. Consumers with longer commutes may pick a longer range EV, and those who barely drive will find long range superfluous. Or, in the other direction, longer range EVs may cause people to choose longer commutes. Regardless, there is potential correlation here that must be addressed.

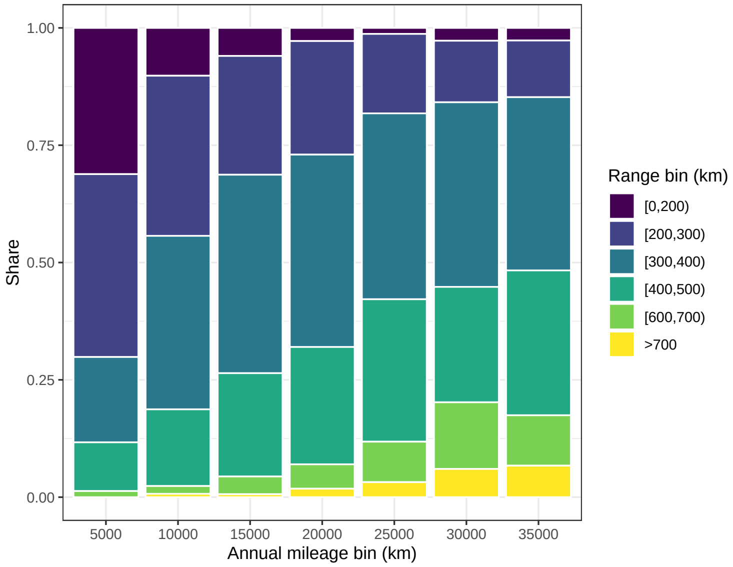


Figure 13: EV range distribution conditional on annual mileage. The data is from the USCALE EV driver survey. The x-axis plots binned mileage (in 5,000 kilometer increments), while the y-axis represents the share of drivers in each mileage level with an EV with the designated range. The color indicates the EV's range (in kilometers), binned in 100km increments.

In fact, I do find evidence suggesting this correlation is important. In the USCALE driver survey data, I observe both the (binned into 100 km intervals) range of the driver's EV as well as their annual mileage (binned into 5,000 km intervals). Figure 13 illustrates the share of survey respondents within each annual mileage bin who select an EV in the specified range bin. Range is indeed correlated with mileage: high mileage drivers are more likely to have long range vehicles than low mileage drivers.

To simulate consumers' choice of EV while preserving this correlation structure, I first use their (simulated) annual mileage together with the survey data to determine the probability they pick each range bin. After picking a range bin, I then randomly draw an EV from within the selected range bin with probability proportional to its market share within that range segment.

Consequently, each consumer is now assigned an EV with corresponding range, battery size, charging speed, and other vehicle characteristics used within the model. Now that I know their EV's range, I can construct the number of charging events needed to fulfill the charging needs implied by their (simulated) mileage. This is obtained simply by taking the incurred mileage and multiplying by their EV's average efficiency (in terms of kWh/km), after which I divide by 0.6 times the size of the battery (corresponding to a charge from 20-80%). This yields the number of (normalized) charging events required by their driving behavior.

Home and work charging availability — A key component of my model is the set of alternative charging options available to the consumer. I simulate these as a function of consumers' region type according to the probabilities in Table 12. I assume that home charging availability is independent of work charging availability. This allows me to sample home and work charging independently of each other.

Home and work charging prices — I simulate the charging prices consumers face at home and the workplace (if available). To understand the distribution of home charging prices, I use data from a large-scale German home charging survey conducted in 2022. This survey involved households who had received a subsidy for the installation of home charging equipment. It asks them about many aspects of their charging experience, but the most important for my purposes is that it asks consumers to report the price per kWh they pay to charge at home. Further details can be found in the appendix.

This survey was conducted in 2022, while my sample is in September of 2024. This means that the average home charging price in the survey (€0.31/kWh) will not accurately reflect the prices consumers face in September 2024. I assume that the mean charging price for consumers in my sample is equal to the mean German household electricity price in the second half of 2024 (€0.39/kWh)[15]. As a necessary assumption, I assume that the standard deviation of prices is the same between both periods. After winsorizing the data, I estimate a standard deviation of €0.052/kWh. I then simulate home charging prices under the assumption they follow a Normal distribution with $p_h \sim N(\mu_h, \sigma_h^2)$, where $\mu_h =$ €0.394/kWh and $\sigma_h =$€0.052/kWh.

It now remains to simulate work charging prices. I do not observe work charging prices directly, but I do observe summary statistics about the difference between work and home charging prices in the USCALE driver survey. Specifically, survey respondents who have charging at both home and work report the (binned) difference between work and home charging prices (20 cents less, 10 cents less, about the same, 10 cents more, 20 cents more).

I model work charging prices as normally distributed $p_w \sim N(\mu_w, \sigma_w^2)$, where μ_w and σ_w^2 are to be estimated. I estimate these using GMM. Specifically, given that home prices are distributed $p_h \sim N(\mu_h, \sigma_h^2)$ and work prices are distributed $p_w \sim N(\mu_w, \sigma_w^2)$ (and assuming independence of home and work prices[16]), their difference is distributed $p_w - p_h \sim N(\mu_w - \mu_h, \sigma_w^2 + \sigma_h^2)$. I create a set of bins corresponding to each bin in the USCALE driver survey (for example, 10 cents less in the sample corresponds to a price difference ranging from -15 to -5 cents, assuming consumers' responses are rounded to the nearest bin). I form one moment for each bin, comparing the implied probability of price differences within each bin to the observed proportion of consumers with that price difference. I then stack these five moments to form the GMM objective function.

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[15] https://ec.europa.eu/eurostat/databrowser/bookmark/5e77ac1a-5efe-46fb-9471-8bd504bdf6f2?lang=en

I estimate that $p_w \sim N(\mu_w, \sigma_w^2)$ with μ_w= €0.368/kWh and σ_w =€0.098/kWh. The lower average cost and higher dispersion are not necessarily surprising. Commercial electricity prices tend to be lower than household prices, which explains the level shift. Additionally, some employers may subsidize work charging for their employees, which would also lower the average cost. The higher variance can be rationalized by the fact that commercial electricity prices are decreasing in annual usage, meaning that heterogeneity in employer size could play a role in the dispersion of work charging prices.

C.2 Charging station characteristics

I use the scraped charger data in addition to the OSM data to construct the charger characteristics I use in estimation. From the charger data, I observe key variables such as the charger's network, number of chargers, and speed of each charger within the station. I use these to construct variables which count the number of chargers within a number of speed bins with cutoffs of 50, 150, 250, and 500 kW. These bins are motivated by the bunching of charging speeds in the data.

I utilize the OSM data to speak to location characteristics. Specifically, I create indicator variables for the presence of various major road types (Autobahn, trunk, primary, secondary) within 1 km of each station. Additionally, I create indicator variables for whether the charger is located at an Autobahn service area or an unmanaged, non-Autobahn rest area. Service and rest areas are located right off the highway, offering a convenient place to stop. Service areas offer a number of amenities (food, restrooms, etc.), whereas rest areas tend to be rudimentary and offer only basic amenities.

Additionally, I construct indicators for the presence of various amenities within 250m of the charging station. Specifically, I include the following amenity types (as reported in the OSM data): fuel, fast food, restaurant, cafe, toilets, motel, hotel, gas, convenience store, supermarket, variety store, greengrocer. Due to the complexity and heterogeneity of amenities, I don't include detailed information about nearby amenities (e.g. I do not model the effect of being by McDonalds vs. Burger King separately). These are intended more as rough proxies for the presence of nearby amenities which may draw consumers.

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[16] For robustness, I repeated the estimation procedure without the assumption of independence. I estimated a small, negative correlation of $\rho = -0.06$, suggesting that there is not much (if any) correlation.

D German home charging survey

The German National Center for Charging Infrastructure (Nationale Leitstelle Ladeinfrastruktur) conducted a survey of recipients of the Federal Ministry of Transport and Digital Infrastructure's KfW 440 home charging subsidy program. The program gave a €900 for the installation of a private charging wallbox. Over 690,000 charging points were installed through the program. 200,000 recipients were randomly selected between September and October of 2022 to complete a survey regarding their home charging behavior, of which 51,893 responded.[17]

The survey asked them about the wallbox installation process (cost, difficulty), property type and location, vehicle ownership and mileage, and charging behavior. Importantly, the survey reports the household's per-kWh charging price. This is very useful for my analysis, as it allows me to incorporate heterogeneity in the value of charging at home into my model.

The distribution of home charging prices reported in the survey are reported in Figure 14. The mean is around €0.31/kWh, while the standard deviation is around €0.06/kWh. This is a significant amount of dispersion. One may think that this variation could be explained by differences in geography or city type. However, Table 15 shows that there is very little difference across region types - rather, most of the dispersion is *within* the region itself.

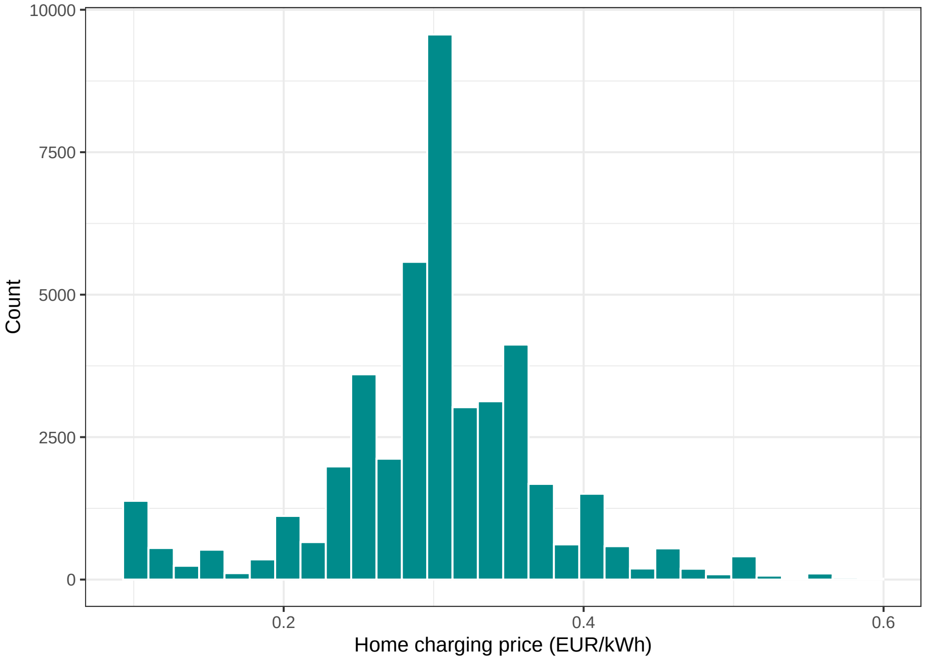

Figure 14: Histogram of home charging prices per kWh (2022)

Why is there so much dispersion in a household's charging price? The short answer is that the German electricity market is one large grid with many public utility companies. The

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[17] https://nationale-leitstelle.de/wp-content/uploads/2024/10/Studie_Einfach_zu_Hause_laden.pdf

Region type	Mean	SD	N
Metropolis	31.08	5.92	2812
Major city, urban	31.21	6.14	4061
Medium city, urban	31.01	6.01	13828
Small town, urban	30.86	6.05	4479
Central rural city	31.21	6.08	1964
Medium rural city	31.06	6.01	6473
Small rural city	31.21	6.18	7961

Table 15: Mean and standard deviation of household electricity prices across region types (2022)

entire market has one wholesale price, rather than location-specific nodal prices like in the United States. While grid fees paid to grid operators can vary across regions, the underlying cost of power is the same.

Additionally, the massive number of public utility companies means that consumers in every market have many provider options. 93% of grid areas have at least 20 providers, and 72% have more than 100 providers.[18] There is a lot of dispersion in the contract offers across providers. Prices per kWh varies based on annual usage levels, energy composition (some offer 100% renewable power), and pricing structure (trading off between a higher upfront fee and the linear price per kWh).

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[18] https://www.bdew.de/media/documents/BDEW_Strompreisanalyse_102025.pdf

E Commercial electricity prices

The most critical input to a fast charger is electricity. In order to model the supply side, I must account for the cost of electricity. The challenge in doing so is that I do not observe the prices that charging networks pay. Instead, I must infer it from available data.

In Germany, the commercial electricity price has several components: the wholesale price, grid fees, surcharges, and taxes. The wholesale price commonly accounts for less than half of the actual price paid by the end consumer. A summary of the electricity prices for German non-household electricity consumers in the second half of 2024 is given in Table 16.[19] In the table, the first column indicates the consumers' binned level of annual usage. The second column lists the average price without any taxes, fees, or other surcharges. The third column lists the average price inclusive of VAT. The fourth column lists the average price inclusive of both VAT as well as mandatory levies and fees.

Quantity bin (MWh)	Price(€/kWh)	Price (€/kWh) (w/VAT)	Price(€/kWh) (w/ VAT, levies, fees)
(0, 20]	0.2820	0.3326	0.3958
(20, 500]	0.2302	0.2714	0.3233
(500, 2000]	0.2010	0.2359	0.2834
(2000, 20000]	0.1810	0.2076	0.2552
(20000, 70000]	0.1509	0.1712	0.2156
(70000, 150000]	0.1313	0.1430	0.1874
>150000	0.1323	0.1390	0.1851

Table 16: Average German commercial electricity price per kWh (in Euros) by electricity usage bin during the second half of 2024. The first column corresponds to the user's annual level of electricity usage. The second column lists the average price per kWh without any taxes, fees, or other surcharges. The third column lists the average price per kWh inclusive of VAT. The fourth column lists the average price per kWh inclusive of both VAT as well as mandatory levies and fees.

From this table, it is evident that the price paid by commercial consumers is decreasing in annual usage. To put these numbers in context of charging network, a small or medium sized network (annual demand between 2 and 20 thousand MWh) would pay 6.78 cents more per kWh (approximately 36% higher) than large networks (annual demand between 70 and 150 thousand MWh).

In my analysis, I will assume that a charging network receives the average electricity price based on its (realized) aggregate usage level. The assumption here is that a charging network will choose a contract with one provider to supply their entire network's charging needs. This is a reasonable assumption given the nature of the German electricity market.

The German electricity market was deregulated in 1996. This spurred significant entry on both the generation and retail side. Today, the market features a large number of retail power providers: in 2023, 93% of grid areas have at least 20 electricity providers, and 72% of grid areas have more than 100 electricity providers [20] The wide variety of retail power providers is facilitated by the fact that the country is treated as a single market with a single wholesale price. As such, a commercial consumer is able to choose a contract with one provider to meet their network-wide electricity needs. This avoids the need to negotiate separate contracts with countless different providers across regions they serve.

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[19] https://www-genesis.destatis.de/datenbank/online/url/3d80e83e

Of course, it is also possible for charging networks to purchase electricity on the futures market to reduce exposure to wholesale price volatility. Additionally, networks sometimes choose to supply power from 100% renewable energy (by sourcing power from suppliers which offer such a mix or using power purchase agreements). Both of these behaviors are unobserved to me, so I abstract away from these considerations in my model.

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[20] https://www.smard.de/page/home/topic-article/211972/212104